Green Chemistry

ROYAL SOCIETY OF CHEMISTRY

TUTORIAL REVIEW

View Article Online
View Journal | View Issue

Check for updates

Cite this: *Green Chem.*, 2023, **25**, 10235

The dawn of aqueous deep eutectic solvents for lignin extraction

Mingyang Hu, Yanyan Yu, Xiaoyan Li, Xinyu Wang and Yun Liu *

Lignin, the most abundant renewable aromatic compound, finds diverse applications across various sectors, including environmental, energy, catalytic, and pharmaceutical industries. The application of aqueous deep eutectic solvents (ADESs) for lignin valorization has garnered significant attention in recent years. However, a comprehensive review addressing lignin extraction using ADESs is currently unavailable, and the implementation of these processes at an industrial scale is still ongoing. This review aims to fill the research gap by presenting recent advancements in lignin valorization using ADESs, with a specific focus on the impact of water on DES properties, such as viscosity, polarity, and nanostructure. The major influencing factors and mechanisms of lignin extraction using ADESs are also highlighted, particularly in optimizing the water content for achieving efficient extraction. Moreover, this review examines the structural features of lignin generated from ADESs and explores its potential valorization. Additionally, an evaluation of the economic and environmental sustainability of lignin extraction using ADESs is provided, highlighting the potential feasibility toward the coming industrial-scale applications. Finally, insights into future perspectives for lignin extraction with ADESs are proposed, reinforcing the status of ADESs as the most promising green solvents for lignin valorization in the coming years.

Received 21st September 2023,
Accepted 15th November 2023

DOI: 10.1039/d3gc03563h

rsc.li/greenchem

Published on 16 November 2023. Downloaded by Westlake University on 11/3/2025 9:33:15 AM.

1. Introduction

It is generally considered that fossil resources can cause severe issues of environmental pollution and resource depletion.[1] To address the energy crisis, it is crucial to develop renewable resources and reduce the reliance on fossil energy sources.[2] Lignocellulose, a renewable alternative resource to fossil, has the potential to be converted into biofuels, high-value chemicals, and biomaterials.[3] By promoting the development and utilization of lignocellulose, we can effectively mitigate the energy crisis caused by fossil energy sources and alleviate the economic challenges associated with it.[4]

Lignocellulose, consisting of cellulose, hemicellulose, and lignin, primarily encompasses agricultural and forestry waste, along with paper industrial by-products.[5] While cellulose and hemicellulose have found industrial applications in biorefining, the extensive utilization of lignin has been hindered by its heterogeneous chemical structure. With a composition of guaiacyl (G), syringyl (S), and *p*-hydroxyphenyl (H) units, lignin comprises 15–35% of lignocellulose and forms complex aromatic polymers through various C–O and C–C bonds (*e.g.*, α-O-4, β-O-4, 4-O-5, β-beta, and β-5).[6] The amount and composition of lignin in lignocellulose vary depending on the plant source or species. Softwoods, for instance, contain about 30% lignin, mainly G units, while hardwoods have 20–25% lignin comprising a mixture of G and S units. Herbaceous species have a minimum of 10–15% lignin, with a diverse composition of G, S, and H units.[7] However, the intricate structure and heterogeneity of lignin pose challenges for its efficient separation, leading to its underutilization.[8] Given that lignin is the second most abundant organic material globally, the effective extraction and valorization of lignin using green and environmental-friendly techniques are imperative in the biomass refining industry.

Deep eutectic solvents (DESs) have gained significant attention as eco-friendly solvents owing to their advantageous properties, such as easy preparation, non-flammability, low toxicity, high thermal and chemical stability, non-volatility, and recyclability.[9] Typically, DESs comprise a hydrogen bond acceptor (HBA) and a hydrogen bond donor (HBD) mixed at the temperatures of 60–80 °C to form a clear solution through hydrogen bonding.[10] Choline chloride (ChCl), betaine (B), glycerol (Gly), urea (U), lactic acid (LA) carboxylic acids, and polyols are common components of DESs.[11] While binary DESs are more prevalent, a few ternary DESs are also reported in the literature.[12] Since their introduction in 2003, various types of DESs have been developed and utilized in biomass treatment and lignin extraction over the past two decades (Fig. 1).[13] Moving forward, the development of economically

Beijing Key Laboratory of Bioprocess, College of Life Science and Technology, Beijing University of Chemical Technology, Beijing 100029, China.
E-mail: liuyun@mail.buct.edu.cn, liuyunprivate@sina.com; Fax: +86-010-64416428; Tel: +86-010-64421335

Published on 16 November 2023. Downloaded by Westlake University on 11/3/2025 9:33:15 AM.

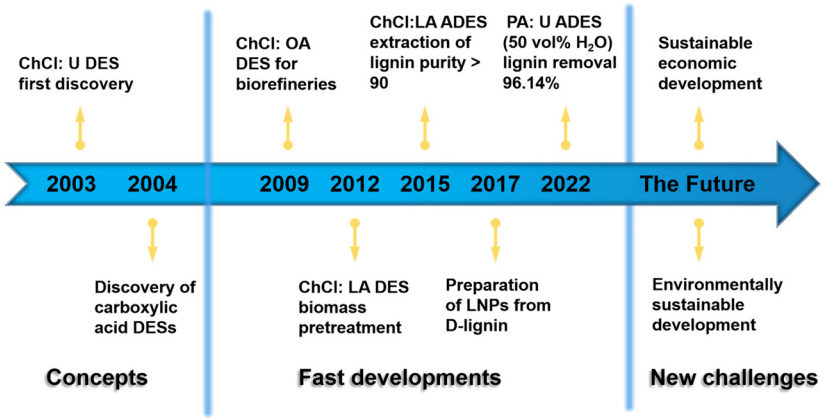


Fig. 1 The milestone history of DESs in lignocellulosic biomass biorefinery.

and environmentally sustainable DESs presents new challenges. Although numerous excellent review articles on DESs along with advancements have been achieved,[14–16] the high viscosity of DESs continues to pose a significant obstacle to their further development.[17] Previous research has shown that the incorporation of water into DESs has led to the creation of aqueous deep eutectic solvents (ADESs), offering a potential solution to reduce the viscosity of traditional DESs and enhance their performance in lignin separation.[18] Nevertheless, the benefits of adding water to DESs for lignin extraction are highly dependent on specific DESs compositions. Moreover, there is a limited and disorganized body of literature on lignin solubilization by DESs in water, with an incomplete understanding of the solubilization mechanisms involved. Therefore, a comprehensive review of studies focusing on lignin extraction using ADESs is necessary. This review should include an in-depth exploration of the role played by ADESs components and provide a clearer understanding of the mechanisms underlying lignin solubilization in ADESs.

To meet the above-mentioned goals, this review initially addresses the impact of water on the properties of DESs, encompassing factors such as viscosity, density, electrical conductivity, polarity, and nanostructure. Subsequently, it provides a quick glance of major affecting factors lignin extraction using ADESs. Additionally, much effort has been made to elucidate the solubilization mechanisms and summarize the structural features of ADES-solubilized lignin, along with highlighting the promising applications of the extracted lignin. Furthermore, the environmental and economic implications of ADESs are assessed. Lastly, the potential perspectives and challenges of lignin extraction using ADESs are proposed.

2. Physicochemical properties of ADESs

As the name suggests, ADESs are formed by adding a certain proportion of water to pure DESs to form a mixture of lower co-solubility without destroying the precursor of the internal hydrogen bonding network structure of DESs. The addition of excess water may lead to the destruction of the hydrogen bonding network and nanostructure in DESs, and the reaction of acidic or basic components with water to form an aqueous solution of HBAs and HBDs. In this case, the special properties of DESs may be lost, and the properties of the solution will be closer to those of conventional aqueous solutions. ADESs, which are critical mixtures between pure DESs and aqueous solutions of HBAs and HBDs, can be improved by the addition of an appropriate amount of water, without destroying the internal network structure of the DESs. As observed in Table 1, the influence of water addition on the physicochemical properties and microstructure of DESs has generated considerable attention, leading to a growing interest in their utilization for the purpose of lignocellulosic fractionation.[19,20] DESs can be categorized as hydrophilic or hydrophobic based on their nature with respect to water.[21] Even a small amount of water can alter the viscosity, conductivity, polarity, and microstructure of DESs.[22] In this section, we will discuss the physicochemical properties of ADES, including viscosity, electrical conductivity, polarity, hydrogen bonding, and nanostructure (Fig. 2).

2.1. Viscosity

The viscosity of ADESs is a crucial physical characteristic, predominantly determined by van der Waals and hydrogen bond interactions.[23] Viscosity plays a vital role in the rate of mass transfer during reactions, where lower viscosity leads to enhanced mass transfer performance and improved separation.[24] However, the high viscosity of DESs poses big challenges in industrial-scale lignin fractionation. Various factors influence the viscosity of DESs, such as the type of HBAs and HBDs, molar ratio, temperature, and water content.[25] Among these factors, water offers significant advantages as a colorless, odorless, non-toxic, and cost-effective solvent to reduce the viscosity of DESs. Research has demonstrated that even small amounts of water can substantially decrease the viscosity of DESs.[24] The introduction of water to DESs leads to alterations in their physicochemical properties, primarily driven by the

 This journal is © The Royal Society of Chemistry 2023

Published on 16 November 2023. Downloaded by Westlake University on 11/3/2025 9:33:15 AM.

Table 1 The summary of physicochemical properties of ADES reported in the literature

DESs composition	HBA : HBD molar ratio	Water content (vol/wt%)	Viscosity (mPa s)	Density (g cm^{-3})	Kamlet–Taft solvatochromic parameters			Ref.
					α	β	π^*	
ChCl : EG		20			1.88	0.81	0.76	27
ChCl : Gly	1 : 2	40	5		0.76	0.76	0.76	42
ChCl : EG	1 : 2	40	4		0.68	0.68	0.68	42
ChCl : BDO	1 : 4	33	6.69	1.05				143
ChCl : PPD	1 : 4	38	9.48	1.05				143
ChCl : Gly	1 : 3	29	13.1	1.18				143
ChCl : Glu	1 : 4	50	8.88	1.30				143
ChCl : M	1 : 3	50	7.92	1.30				143
ChCl : LA	1 : 3	17	12.2	1.16				143
ChCl : MA	1 : 1	11.62	445.9	1.2303				144
ChCl : Gly	1 : 2	5.26	51.3	1.1742				144
βA : MA	1 : 1	19.48	174.6	1.352				144
MA : P	1 : 1	17.81	251	1.3184				144
ChCl : F	1.5 : 1	7.84	280.8	1.2078				144
ChCl : X	2 : 1	7.74	308.3	1.2095				144
ChCl : S	4 : 1	7.4	581	1.2269				144
F : Glu : S	1 : 1 : 1	22	720	1.3657				144
ChCl : Glu	1.5 : 1	7.84	397.4	1.2069				144
ChCl : PPD	1 : 1	7.7	33	1.0833				144
LA : Glu	5 : 1	7.89	37	1.2497				144
ChCl : Sb	1.5 : 1	9.23	138.4	1.1854				144
ChCl : Xl	2 : 1	11.17	86.1	1.17841				144

intense hydrogen bonding interactions occurring between the HBA and HBD present in the DESs components and water molecules. This alteration can positively impact the ability of DESs to dissolve lignin. As the water content increases, it competes with the HBA and binds to the HBD through hydrogen bonding. Consequently, the substituted HBA is present as a cation, promoting the ion dissociation of DESs and ultimately reducing their viscosity.[26] For instance, in the ChCl : ethylene glycol (EG) DES system, the addition of 20 wt% water led to a decrease in the viscosity from 39.4 to 12.0 mPa s.[27] Furthermore, incorporating a 2 wt% NaOH solution into the DES system demonstrated an even greater reduction in viscosity, dropping from 81.0 to 22.8 mPa s. However, the viscosity increased when more NaOH was added. This suggests that water deposited in DESs can modify their viscosity, consequently impacting the pretreatment effectiveness of biomass (Fig. 3).[28]

2.2. Density

Density is indeed another important parameter of DESs.[29] In general, with the exception of hydrophobic DESs, the density of most DESs is higher than that of water, which is typically about 1.2 g cm^{-3} at room temperature.[30] The density of DESs is influenced by several factors. One key factor is the number of hydroxyl groups present in the DESs as the density tends to increase with a higher number of hydroxyl groups. On the other hand, the density decreases with the length of the hydrophobic alkyl chain in the HBDs component of the DESs.[16] By varying the composition and molar ratio of the HBA : HBD, the density of DESs can be modified. Alcalde *et al.* conducted a study employing molecular dynamics (MD) simulations to predict the densities of DESs with varying different molar ratios.[31] In the case of ChCl : LA ADES, the density values increased with an increasing molar ratio of HBA to HBD. Additionally, temperature has an impact on the density of DESs, typically resulting in a gradual decrease as the temperature increases.[32] When water is added to DESs, it can interact *via* hydrogen bonding with the HBD and HBA components. As a result, the original hydrogen bonding network in the DES is partially disrupted, leading to a decrease in the density of the DESs.[24] For example, the density of pure ChCl : U DESs was measured to be 1219.4 kg m^{-3}, but it decreased to 1111.3 kg m^{-3} upon the addition of 50 vol% water.[18] The densities also varied nonlinearly with increasing water concentration. The addition of water caused a moderate volume expansion of ChCl : LA DES, and this expansion increased linearly with the concentration of water. This expansion suggests that water molecules tend to solvate DESs, isolated HBA : HBD clusters.

Therefore, the density of DESs can be adjusted by tuning the composition and molar ratio of HBA to HBD, temperature, and the addition of water. Better understanding the density behavior of DESs is crucial for optimizing their applications, particularly in the process of biomass pretreatment.

2.3. Electrical conductivity

The conductivity of DESs primarily depends on the viscosity of the solvent. Higher viscosity leads to weaker conductivity, resulting in low ionic conductivity of DESs.[33] The high conductivity of DESs can provide more ions during the pretreatment process, accelerating the rate of the lignin depolymerization reaction. In addition, the conductivity of the solution can be increased, resulting in a more uniform distribution of the reac-

Published on 16 November 2023. Downloaded by Westlake University on 11/3/2025 9:33:15 AM.

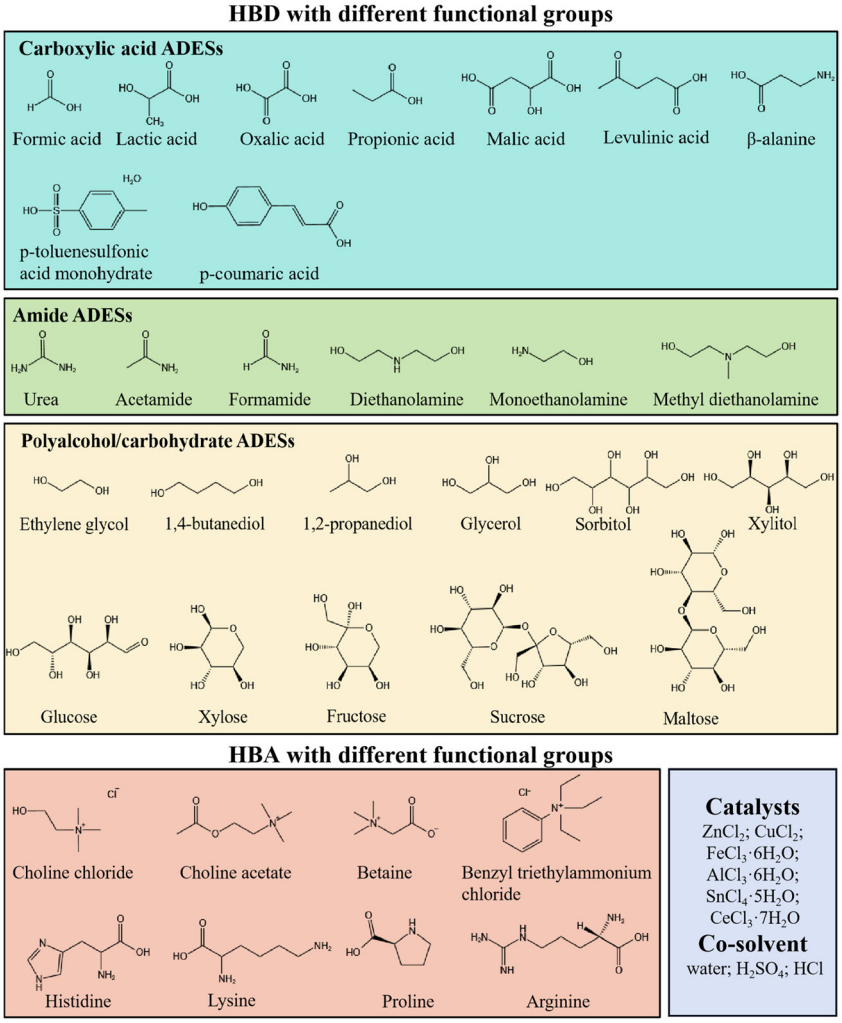


Fig. 2 The representative structures of HBDs and HBAs of ADESs along with catalysts and cosolvents.

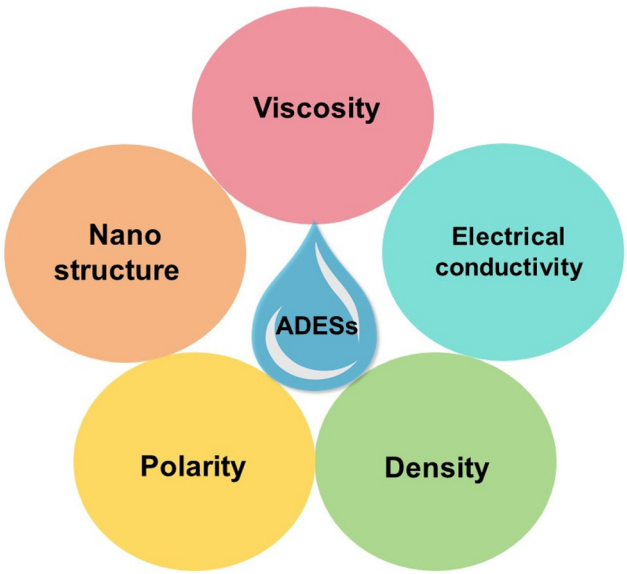


Fig. 3 The properties description of ADESs.

tants in the solution. This helps to improve the homogeneity of the reaction and avoid localized reactions. In addition, the high electrical conductivity improves the solute solubility in the solvent, thereby increasing the solvent recovery. It has been demonstrated that the conductivity of DESs is also influenced by the nature of HBA and HBD, molar ratio, and water content added.[34] As the water content increases, the conductivity also increases. Typically, ADESs demonstrate higher conductivity compared to equivalent DESs without water. This can be attributed to the increase in charge ion carriers and a shift toward electrolyte-like solution behavior.[35] Moreover, the presence of numerous hydrogen bonds in water enhances the mutual solubility with DESs, promoting the ion dissociation of DESs components through enhanced hydrogen bonding interactions.[22] The conductivity reaches its maximum when the water content in DESs reaches 60% but decreases upon further water addition.[22] This decrease is due to electrolyte dilution caused by excessive water addition, resulting in decreased conductivity. In addition, other factors such as microwave irradiation and high temperature can enhance DESs conduc-

 This journal is © The Royal Society of Chemistry 2023

Published on 16 November 2023. Downloaded by Westlake University on 11/3/2025 9:33:15 AM.

tivity through molecular collisions. High DESs conductivity offers advantages for lignin fractionation.

2.4. Nanostructure

The DESs consist of disorganized networks of molecular ion clusters held together by hydrogen bonding interactions between HBA and HBD.[36] Incorporating water in DESs can probably adjust and modify the DESs nanostructure.[37] Various techniques, such as neutron scattering, empirical potential structure refinement (EPSR), ^{1}H-NMR, Fourier transform infrared spectroscopy (FT-IR), and Raman spectroscopy, enable a thorough investigation of the influence of water on the nanostructure of DESs.[22] Structural analysis reveals that as water is added, the hydrogen bonds gradually break, but the overall supramolecular structure of DESs is maintained until the water content reaches 40%–50%. Upon the addition of additional water, the HBA and HBD present in DESs undergo complete separation and hydration, resulting in the formation of distinct aqueous solutions of HBA and HBD. Hammond *et al.* conducted a comprehensive characterization of the nanostructure of ChCl : U ADES using total neutron scattering and EPSR.[38] Their findings indicated that at water contents below 6.5 wt%, water had a lesser impact on disrupting the DESs hydrogen bonding network and, conversely, strengthened the ChCl : U hydrogen bonding. Between 12.2 wt% and 41 wt%, DESs clusters remained in their current state but were separated by water. Once the water content reaches 51 wt%, a significant change occurred in the solvation process, resulting in a reduced prevalence of DESs structures and the formation of a molecular-level aqueous solution known as a DESs component. The nanostructure of DESs exhibits minimal variations for different water contents, as observed in three-dimensional spatial density function (SDF) plots.[38] From the analysis of Fig. 4, it was observed that U–Cl$^-$ binding to choline could still occur in DES systems with high water content. However, the interactions between choline and water, as well as between urea and water, became dominant, causing the ChCl : U DES–water interactions to weaken. As the water content increases, the hydration of the DES components also increased, reaching a saturated first hydration shell for urea at a water content of 51 wt%, similar to the situation with choline. The interactions between water and Cl$^-$ progressively increased with the increase in hydration, and the presence of a second solvation shell for water indicated that water-Cl$^-$ interactions were stronger compared to choline–Cl$^-$ and U–Cl$^-$ interactions. The solvation of water by Cl$^-$ further suggested a transition from DESs to aqueous solutions. Therefore, the decomposition of the DES structure correlates with the point at which the DES–water interaction becomes dominant over the DES–DES interaction.

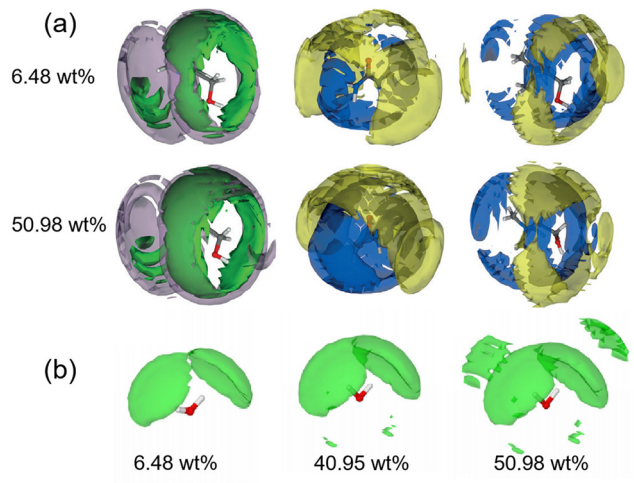


Fig. 4 The SDF plots of 3D nanostructures of DESs with different water contents. (a) Center molecules are choline (left), urea (center), and choline (right). (b) Intermediate molecules are water molecules; chloride (green), urea (lavender), choline (yellow), and water (blue) molecules (reproduced from ref. 38. Copyright (2017) Wiley).

Sapir *et al.* conducted MD simulations to investigate the influence of water on the nanostructure of ChCl : U DES.[39] The simulation findings indicated that even at low hydration levels, water had an effect on altering the nanostructure of DESs. As the water content increased, the intermolecular distances remained unchanged, but there was a gradual hydration of HBA and HBD. In the study by Gabriele *et al.*, the effect of water addition on DESs nanostructure of ChCl and EG was investigated using FT-IR and NMR techniques.[22] The results indicated that when water was added to the DESs, the interaction between ChCl and EG gradually weakened until the water content reached 50% (w/w). However, even with this dilution, the supramolecular structure of the DESs was partially retained. This suggested that the DESs underwent a gradual disruption of their nanostructure with increasing water content. Upon further addition of water, beyond the 50% threshold, the DESs component was completely dissociated and hydrated. This meant that the ChCl and EG components lost their supramolecular arrangement and formed separate entities in the presence of excess water.

2.5. Polarity

The polarity characteristics of ADESs can affect the efficiency and selectivity of delignification by influencing the solvation capacity, ionization capacity, and solvent selectivity.[40] The Kamlet–Taft solvatochromic parameters, namely, α, β, and π^*, are commonly utilized to describe these characteristics.[15] These parameters represent the hydrogen-bonding acidity, hydrogen-bonding alkalinity, and polarizability of ADESs, respectively.[41] Typically, the Kamlet–Taft solvent color parameters can predict the delignification effect of ADESs. It was demonstrated that the α and β values had a stronger correlation with lignin solubility compared to the π^* values.[42] Surfaces with higher α values, which indicate higher hydrogen-bonding acidity, can effectively separate acidic protons from ADESs and break the LCC bond.[15] When synthesizing ADESs, the compositions of HBA and HBD, as well as water content are crucial in determining these parameters. Li *et al.* revealed that in alkali-enhanced ChCl : EG DES, the initial α, β, α–β, and π^* values were 1.45, 0.7, 0.75, and 0.87, respectively.

Published on 16 November 2023. Downloaded by Westlake University on 11/3/2025 9:33:15 AM.

However, with the addition of 20% water, these parameters increased significantly to 1.88, 0.81, 1.08, and 0.76, leading to a substantial improvement in lignin removal.[27]

3. Extraction of lignin by ADESs

DESs have emerged as a promising green environmental-friendly solvent for the extraction of lignin.[43] In comparison to cellulose and hemicellulose, DESs exhibit favorable dissolution and regeneration properties for lignin. It is primarily due to the rich hydrogen-bonding network structure present in DESs, enabling strong interactions with lignin molecules.[76] On the other hand, the intramolecular/intermolecular multiple hydrogen-bonding network structure of cellulose and its highly oriented supramolecular crystalline structure pose significant limitations to cellulose solubilization in DESs.[14,44] Table 2 provides information on lignin removal by different types of ADESs and the effect of water content on the removal efficiency. It offers valuable insights into the effectiveness of various ADESs systems for lignin extraction. In this section, lignin extraction using different types of ADESs was addressed, revealing the advantages and disadvantages of each ADESs for delignification.

3.1 Extraction of lignin with carboxylic acid ADESs

Carboxylic acid DESs, such as those based on LA, oxalic acid (OA), and formic acid (FA) as HBD, and ChCl and B as HBA, have shown promising results in lignin extraction. In ChCl : LA ADES, the degradation rate and removal efficiency of lignin can vary depending on the water content in the DES.[45] For instance, in Fig. 5, it was shown that the removal rate of lignin in ChCl : LA ADES was 35% when the water content was 10 wt%, 87.59% at 20 vol%, and 20.64% at 50 wt% water content. When rice straw was pretreated using ChCl : LA DES, the addition of 5.0% (v/v) water resulted in an extraction efficiency of lignin as high as 90%.[46]

The efficacy of ADESs in solubilizing lignin is a primary factor that determines their effectiveness of lignin extraction from biomass. Soares *et al.* synthesized 22 kinds of ADES using different combinations of organic acids, sugars, and polyols as HBDs, and ChCl and urea as HBAs. They investigated the solubility of lignin monomer model compounds (ferulic, syringic, vanillic acids, and syringaldehyde) and industrial lignin in these ADESs.[47] The results indicated that PA : U ADES exhibited the highest solubility among the tested ADESs, with a significant increase in the solubility for the lignin model compounds. The synergetic effect of the HBD and HBA components contributed to the increased solubility of lignin model compounds. Moreover, the presence of 50 wt% water in PA : U ADES increased the solubility of kraft lignin by 1181.7 ± 29.2-fold and organic solvent lignin by 228.5 ± 9.7-fold. The solubility mechanism of lignin in ADESs is suggested to be hydrophilic, as indicated by dynamic light scattering (DLS) analysis, which showed the dominance of hydrophilic solvation aggregates for aromatic and phenolic acids produced during lignin depolymerization. Notably, it's important to mention that adding excessive water to ADESs might adversely affect their structure, leading to a potential decrease in the delignification efficiency. Zhang *et al.* synthesized four DESs using L-arginine (L-Arg) and L-proline (L-Pro) as HBAs, and EG and Gly as HBDs, comparing them with ChCl-DESs for lignin solubilization.[48] Results showed that DESs with higher α values had more favorable lignin dissolution due to the

Table 2 Conditions for lignin extraction from biomass using ADES

DESs (HBA : HBD)	Biomass	Water content (vol/wt%)	Conditions: temperature (°C), time (h), and solid loading (wt%)	Lignin removal (%)	Ref.
Carboxylic acid ADESs					
ChCl : LA	Eucalyptus	20	140 °C, 4 h, 1 : 6	87.59	53
ChCl : LA	Rice straw	10	120 °C, 3 h, 1 : 20	64.8	145
ChCl : FA	Napier grass	50	120 °C, 1.5 h, 1 : 10	42.17	51
ChCl : OA	Oil palm fronds	16.4	120 °C, 45 min, —	74.7	146
B : Lys	Corncobs	87	60 °C, 5 h, 1 : 20	49.06	50
ChCl : GA	Hybrid poplar chips		140 °C, 3 h, 1 : 20	50	95
BTEAC : FA	Bamboo shoot shells	10	130 °C, 20 min, 1 : 10	88.6	40
B : LA	Rice straw	10	60 °C, 12 h, 1 : 10	57.77	46
Amide ADESs					
CL : U	*Populus tomentosa*	5	90 °C, 6 h, 1 : 20	36.8	147
ChCl : U	Oil palm frond	2.5	Ultrasonic pretreatment 40 min, 2.5% NaOH, 120 °C, 4 h, 1 : 10	47	65
ChCl : U	*Eucalyptus globulus* Labill.	50	90 °C, 8 h, 1 : 20	80.64	49
Polyalcohol ADESs					
ChCl : EG	*Populus deltoides*	20	140 °C, 3 h, 1 : 10	62.19	27
ChCl : EG	Switchgrass	20	130 °C, 0.5 h, 1 : 5	80	57
ChCl : EG	Sugarcane bagasse	20	100 °C, 0.5 h, 1 : 10	84	42
ChCl : BDO	Moso bamboo	10	110 °C, 1 h, 1 : 2	63.1	56
ChCl : Gly	Switchgrass	20	120 °C, 1 h, 1 : 10	63.84	79
ChCl : Gly	Corn stover	20	120 °C, 4 h, 1 : 10	37.9	148
ChCl : Gly	Corn straw	20	120 °C, 3 h, 1 : 20	57.87	149
ChCl : PCA	*Albiziae Cortex*	5	180 °C, 7 h, 1 : 10	64.03	150

 This journal is © The Royal Society of Chemistry 2023

Published on 16 November 2023. Downloaded by Westlake University on 11/3/2025 9:33:15 AM.

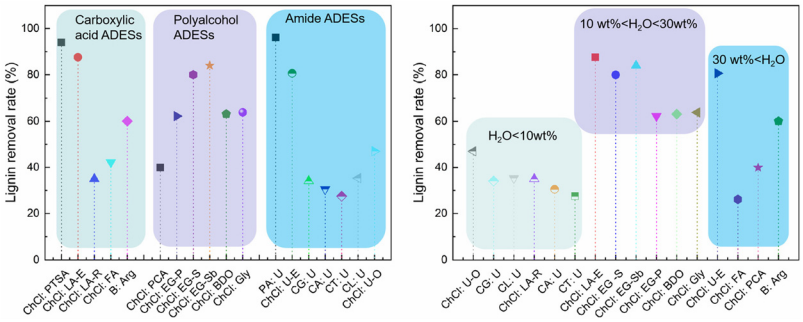


Fig. 5 The fractionation of lignin by different types of ADESs. (a) The effect of different HBDs in ADESs and (b) the effect of different water content in ADESs. The ChCl : LA-E : ChCl : LA ADES extraction of *Eucalyptus* lignin. ChCl : LA-R : ChCl : LA ADES-extracted rice straw lignin. ChCl : EG-P : ChCl : EG ADES-extracted *Populus deltoides* lignin. ChCl : EG-S : ChCl : EG ADES-extracted switchgrass lignin. The ChCl : EG-Sb : ChCl : LA ADES extraction of sugarcane bagasse lignin.

lignin's high content of β-O-4 bonds. The highest α value (2.35) and best lignin solubility (29.09 wt%) were achieved with ChCl : LA ADES (2 wt% H_2O). The added water significantly enhanced mass transfer and activated reaction sites, promoting lignin solubilization in ADES. Compared to L-Arg : EG ADES (2 wt% H_2O), solubility increased 3.49 and 2.16 times at concentrations of L-Arg : EG ADES (70 and 90 wt% H_2O), respectively. However, the addition of water was found to have a disadvantage for ChCl-based DESs. It was due to water's tendency to interact more strongly with the more electronegative Cl^-, resulting in a stronger interaction energy between HBD and Ch^+.

Various DESs exhibit varying outcomes in lignin extraction from biomass, even when subjected to an equal quantity of water addition. In the study by Soares *et al.*, they investigated four DES (PA : U, ChCl : U, ChCl : LA, ChCl : PA DES) with the addition of 50 wt% water for lignin extraction from *Eucalyptus globulus Labill.*[49] The most effective DES was found to be PA : U ADES, which treated the sample at 90 °C for 8 h with 25% organic acid (PTSA), resulting in 96.14% lignin removal. Liang *et al.* synthesized three high water content (65%–93%) ADES using B as the HBA and three amino acids (lysine, arginine, and histidine) as the HBDs to extract lignin from corn kernel species.[50] They added 65%–93 wt% water to the DESs and found that B: lysine (Lys) ADES exhibited the highest solubility of lignin. With the addition of 87 wt% water, B: Lys ADES achieved the highest lignin solubility. On the other hand, B: arginine (Arg) ADES with 73 wt% water was the most effective in removing 60.02% of lignin from corn kernels. Gundupalli *et al.* investigated the use of ChCl : FA and ChCl : LA ADES for the delignification of *Napier* grass at 120 °C for 90 min with a solids loading of 10% w/w.[51] The results showed that the removal of lignin with ChCl : FA ADES was 26.13%, while it was 20.65% for ChCl : LA ADES. The percentages were significantly lower compared to the DESs without water addition, where the removal of lignin was 42.17% and 57.37%, respectively. At water contents exceeding 30%, the effectiveness of DESs in lignin extraction decreases, resulting in reduced solubility of lignin. However, blending ChCl : FA and ChCl : LA DESs leads to 61.76% delignification as they compete with lignin for interaction by forming intermolecular hydrogen bonds.

Balasubramanian *et al.* synthesized ADES using renewable *p*-toluene sulfonic acid (*p*-SA) as HBD for lignin extraction from rice husks, achieving 83% yield.[52] Controlling the water content of ChCl : *p*-SA ADES within 26%, they found that adding water had a positive effect on lignin yield, reaching 94% at 5.3% water content but dropping to 47.9% when water was further increased to 26%. This phenomenon was probably attributed to water's influence on the physicochemical properties of DES.

Cui *et al.* used ChCl : LA ADES to extract lignin from eucalyptus and used the remainder to prepare pulp.[53] The addition of 20 vol% water to ChCl : LA DES resulted in an 87.59% of lignin removal at 140 °C, with a maximum pulp yield of 60.11%. Utilizing water precipitation, it was possible to obtain 80.11% of lignin from logs with high purity, a uniform molecular weight distribution, and low condensation. The use of ChCl : LA ADES offers advantages in terms of reducing production costs and obtaining high-quality products. It demonstrated the potential for ChCl : LA ADES to serve as an ideal industrial solvent, promoting environmental sustainability and economic efficiency in lignin extraction. Hou *et al.* conducted a study on the two-stage extraction of lignin from rice straw using ADESs.[54] In the first stage, they treated rice straw with malic acid : proline (MA : P ADES) or ChCl : OA ADES, and in the second stage, they used ChCl : U ADES. The addition of 5% water to each ADES species was employed to regulate the severity of the treatment process. The results showed a significant increase in lignin removal up to 76.82% with the combined treatment of MA : P ADES and ChCl : U ADES.

3.2 Extraction of lignin with polyhydric alcohols ADESs

Polyhydric alcohol-based ADESs are synthesized ADESs using polyols like Gly, EG as HBDs for delignification.[55] In polyhydric alcohol DESs, the hydroxyl groups of the polyols can interact with the ether bonds present in the lignin structure, facilitating the delignification process.[15] The chemical structure of

Published on 16 November 2023. Downloaded by Westlake University on 11/3/2025 9:33:15 AM.

the polyols plays a significant role in determining the effectiveness of the polyhydric alcohol DESs in lignin removal. Currently, the commonly used polyol DESs are predominantly basic DESs, with Gly, EG, and 1,4-butanediol (BDO) as the primary HBDs. Cheng *et al.* conducted research on the delignification of moso bamboo using a DES synthesized from ChCl : BDO DES.[56] They varied the moisture content of the bamboo samples at 60%, 50%, 40%, 30%, 20%, and 10%. The highest lignin removal achieved was 63.1% after 1 h of reaction at 110 °C, with a solid–liquid ratio of 1 : 2 and a moisture content of 20%. Higher water content in the solids resulted in an increased recovery of solid residue from 62.4% to 78.6% while weakening the delignification efficiency. This phenomenon could be explained by the disruption of the hydrogen bonding network in DESs due to the presence of moisture in the feedstock, which is similar to the principle observed in ADESs processes.

Polyol-based ADESs typically exhibit a neutral or slightly alkaline nature. To modulate the pH of these ADESs, NaOH, H_2SO_4, and HCl were added, thereby facilitating the regulation of the Kamlet–Taft solvatochromic parameters. Consequently, this pH adjustment contributed to enhancing the efficacy of lignin removal. Another study by Chen *et al.* investigated the delignification of ChCl : EG ADES with a solid loading of 20 wt% at 130 °C and acidic conditions for 30 min. The results showed 80% lignin removal from switchgrass. The presence of 10% and 20% water did not significantly affect the fractionation performance of ChCl : EG ADES when switchgrass had a solid loading of 20%. However, at 27% solids loading, 10% and 20% water had a lesser effect on delignification.[57] The findings highlight the potential of polyhydric alcohol ADES, particularly ChCl : EG ADES with water, in lignocellulosic biomass pretreatment for efficient delignification. Li *et al.* reported on an alkalinity-enhanced ADES for the delignification of poplar wood. The study investigated ChCl : EG ADES containing 20% water and 3 wt% NaOH at 140 °C for 3 h, which resulted in 62.19% lignin removal.[27] It was found that lignin removal was positively correlated with the difference in hydrogen-bonded acidity and alkalinity (α–β) of the ADES. ChCl : EG ADES had an α–β value of 1.08, which was greater than that of other DESs with different alkali concentrations or water contents. Additionally, the solvent bipolarness (π^*) of ChCl : EG ADES was lower compared to other DES types, indicating its effectiveness in lignin removal from poplar wood. The study yielded 43.00 kg of fermentable sugars and 12.98 kg of lignin, which contained rich β-O-4 bonds, from 100 kg of poplar wood. Chourasia *et al.* synthesized ternary ADES for the delignification of sugarcane bagasse using $MgCl_2{\cdot}6H_2O$ and $NiCl_2{\cdot}6H_2O$ as Lewis acids, ChCl as HBA, EG, and Gly as HBD.[42] Their findings revealed that ChCl : EG : NI ADES (20% H_2O) with 10% solid loading, treated at 100 °C for 30 min with the assistance of microwaves, achieved an 84% lignin removal. This result surpassed the performance of ADES with 40% water content and pure ChCl : EG ADES. Water was also utilized as an antisolvent to enhance lignin solubility, resulting in the extraction of 26% lignin from ADES.

3.3 Extraction of lignin with amide ADESs

The amide-based DESs system has recently received more and more attention in comparison with carboxylic acid-based and polyols DESs.[58] It could be attributed to the superior performance of alkaline DESs in lignin extraction. Within an alkaline environment, the ether bond in lignin and the ester bond between hemicellulose and lignin are more easily broken, resulting in significant lignin removal.[16] However, acid-based ADESs can not only partially remove lignin but also remove the acid-sensitive hemicellulose fraction. As a result, although amide-based DESs exhibit higher lignin removal capacity, acid-based DESs enhance the enzymatic efficiency of lignocellulose to a greater extent. To address the challenges associated with DESs in biomass conversion and stabilization, New *et al.* reported the use of ChCl : U ADES for the delignification of oil palm fronds (OPF) by introducing 30 vol% of water into the DES.[18] The results indicated that OPF lignin removal reached 17.11% when treated at 120 °C with a solid–liquid ratio of 1 : 10 (w/v) for 4 h. However, a further increase in water content resulted in a decrease in lignin removal, suggesting a weaker delignification capacity when introducing a larger proportion of water into the DES.

Zhang *et al.* conducted a study where they synthesized four DESs for lignin solubilization, comparing them to the solvents derived from ChCl. The DESs were formulated using L-arginine (L-Arg) and L-proline (L-Pro) as HBAs, and glycerol as HBDs.[48] The results demonstrated that the DESs synthesized using L-Arg and L-Pro exhibited higher hydrogen bond providing capacity (represented by the α value) due to the higher content of β-O-4 bonds in lignin. This higher α value favored the dissolution of lignin in these DESs. Among the ChCl : LA DESs with 2 wt% water, the highest α value (2.35) and the best lignin solubility (29.09 wt% water) were observed. Various characterization techniques revealed that the addition of water significantly increased the mass transfer capacity of L-Pro and L-Arg-based DESs, thus activating the reaction sites and promoting lignin solubilization in these DESs. Specifically, the solubility of lignin in L-Arg : EG DES (70 and 90 wt% water) concentrations was increased by a factor of 3.49 and 2.16, respectively, compared to L-Arg : EG DES with 2 wt% water. However, it was also noted that the addition of water had a negative impact on ChCl-based DESs. This was attributed to water having a stronger interaction with the more electronegative Cl^- ion present in ChCl, resulting in increased interaction energy between the HBD and Ch^+. This stronger interaction hindered the solubilization of lignin in ChCl-based DESs. The findings suggested that carefully designing DESs with appropriate components and water content can enhance lignin extraction.

4. The affecting factors and mechanisms of lignin extracted by ADESs

The lignin extraction from lignocellulose using ADESs has shown promising results and has received considerable atten-

 This journal is © The Royal Society of Chemistry 2023

Published on 16 November 2023. Downloaded by Westlake University on 11/3/2025 9:33:15 AM.

tion. However, the complexity and heterogeneity of lignin make it challenging to determine the extraction mechanisms.[59] Lignin extraction from biomass using ADESs can generally be categorized into two main processes: hemicellulose hydrolysis and lignin extraction. Hemicellulose hydrolysis involves the cleavage of ether and ester bonds present in the LCC. This process is often accompanied by depolymerization and polymerization reactions of lignin, making the precise reaction mechanisms unclear.[60] Regarding lignin extraction by ADESs, several factors influence the lignin extraction efficiency, including the composition of the HBD and HBA, water content, temperature, reaction time, and molar ratio. Understanding the interactions between ADESs components and lignin, as well as the major effecting factors, is essential for designing efficient lignin extraction processes using ADES. Henceforth, this section offers a comprehensive overview of the intricate mechanisms entailed in the extraction and isolation of lignin from lignocellulose using ADESs, which are subject to diverse influencing factors.

4.1. Effect of HBD on lignin extraction

HBD plays a vital role in lignin separation, encompassing urea, thiourea, carboxylic acids, polyols, amino acids, sugars, and other acidic or basic reagents.[61] Thus, investigating the structural changes of lignin in acidic and basic ADESs is crucial for understanding the mechanism behind lignin extraction using ADESs. The current primary mechanism for lignin depolymerization involves cleaving the β-O-4 bond between the phenylpropane units.[62] Fig. 6(a) illustrates three main reaction pathways for β-O-4 bond cleavage in acidic ADESs.[15] Pathway I involves mild acid-catalyzed conditions where the α-OH group undergoes dehydration, leading to protonation and the formation of a stable benzyl carbon cation. Thereafter, the benzyl carbon cation eliminates the terminal γ-CH_2OH group, resulting in the production of α–β unsaturated compounds and formaldehyde, which further facilitates β-O-4 bond breaking. Pathway II occurs when the α-OH group is protonated to create a stable benzyl carbon cation, followed by the formation of an enol ether intermediate through the elimination reaction of the α-ligand and β-H. Subsequent hydrolysis leads to β-O-4 bond cleavage and the formation of a lignin-bound Hibbert ketone (LBHK) structure, ultimately yielding depolymerized compounds like mono- or diketones and lignin-stabilized structural units.[63] Pathway III involves the allylic rearrangement of the LBHK structure, followed by hydrolysis, resulting in β-O-4′ bond cleavage and the generation of mono- or diketones and

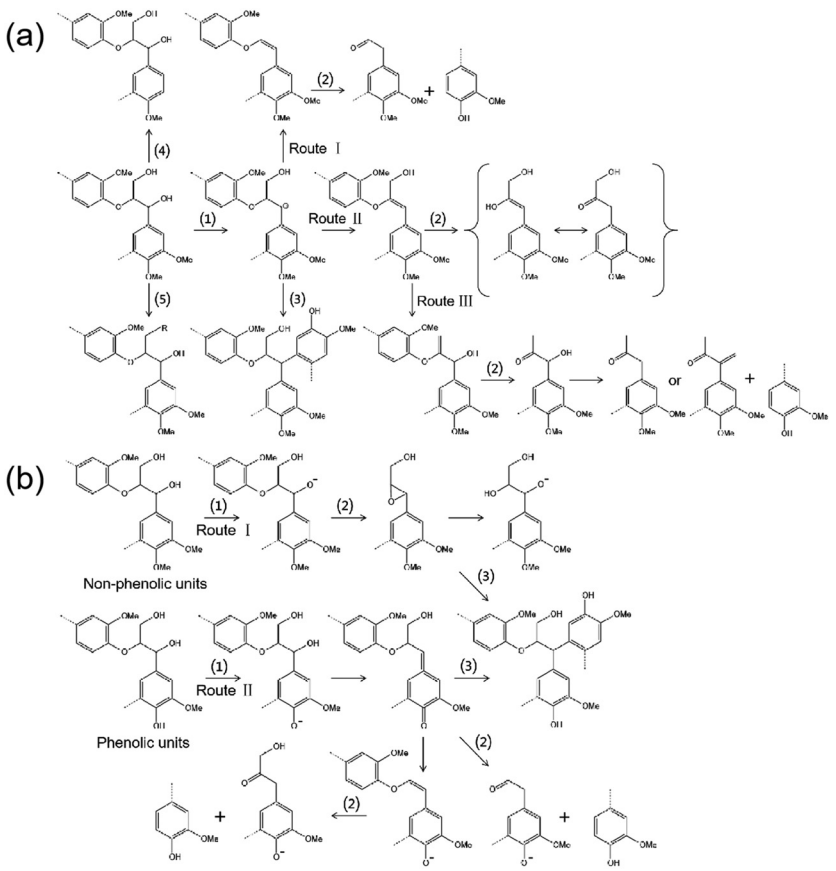


Fig. 6 (a) The reaction mechanism for the lignin extraction by acidic ADESs. (b) The reaction mechanism for the lignin extraction by basic ADESs. (1) Dehydration' (2) depolymerization, (3) repolymerization, (4) acylation, and (5) demethoxylation.

Published on 16 November 2023. Downloaded by Westlake University on 11/3/2025 9:33:15 AM.

lignin-stabilized structural units. Moreover, acidic ADES can break certain C–C bonds (*e.g.*, β-5′ and β–β′) in lignin, facilitating the separation of lignin from lignocellulose.[64]

The inclusion of polar groups, such as hydroxyl groups, within lignin gives rise to robust intermolecular forces, rendering lignin insoluble in aqueous solutions. However, the deprotonation of the phenol OH group renders lignin soluble in alkaline aqueous solutions. Consequently, alkaline conditions are commonly employed in the extraction of lignin from lignocellulose, particularly through alkaline ADESs, such as ChCl : U ADES, which can disrupt LCC interactions and cleave β-O-4 bonds.[65] The mechanism by which alkaline ADESs separate lignin and cleave the β-O-4 bond is depicted in Fig. 6(b). The reactivity of β-O-4 bonds under alkaline conditions is dependent on phenolic and non-phenolic units. β-O-4 bond cleavage in non-phenolic units results in the formation of epoxide intermediates, which exhibit relatively slow reactivity. In contrast, phenolic units are more susceptible to depolymerization, and the deprotonation of phenol OH leads to the formation of a quinone-methyl type intermediate. At this stage, nucleophilic reactions are likely to occur, followed by various depolymerizations and polymerizations.

4.2. Effect of HBA on lignin extraction

HBAs, such as ChCl and B, play crucial roles in lignin extraction performance. DESs outperform individual acids or other HBD in terms of lignin removal. For instance, the lignin yield with pure LA was 41.52%, while ChCl : LA ADES could achieve a lignin yield of approximately 70%.[66] ChCl, the most commonly used HBAs in ADES, exhibits superior lignin separation compared to other HBAs. This can be attributed to Cl^- ions in ChCl-based ADESs, which interact effectively with cellulose and hemicellulose components in lignocellulose.[53] Cl^- ions form strong hydrogen bonds with lignin, promoting the stretching and breaking of the LCC bond, thus facilitating lignin solubilization.[15] Therefore, the delignification efficiency relies on the ability of electronegative halogen anions in DESs to form hydrogen bonds with lignin. Cl^- ions have been demonstrated to facilitate the cleavage of the β-O-4 bond and hinder lignin condensation.

Using 2-phenoxy-1-phenylethanol (PPE) as a lignin model compound,[67] the mechanistic steps involved in the transformation of PPE using ChCl-based DESs is illustrated in Fig. 7. These steps include (i) hydroxyl protonation, (ii) nucleophilic attack of the halide anion on the α carbon, resulting in protonation of the hydroxyl group, (iii) subsequent nucleophilic attack of the halogenated anion on the α carbon, causing the dehydration and release of the carbonyl ion, (iv) formation of a halogenated PPE intermediate, (v) dehydrochlorination leading to the formation of an enol-ether intermediate, and finally (vi) cleavage of the ether bond. PPE exhibits rapid dehydration, generating stabilized intermediates that decrease the energy required for ether bond cleavage, leading to a lower ΔG.

4.3. Effect of water content on lignin extraction

The water content in ADESs plays a significant role in lignin separation and extraction. The addition of a certain amount of water to DESs reduces their viscosity, making them more fluid and overcoming the high viscosity disadvantage. In a study by Cui *et al.*, ChCl : LA ADES with 20% water addition demonstrated lower viscosity and density, resulting in lignin removal exceeding 85%.[53] The reaction mechanism of lignin extraction using ADESs affecting the water content is depicted in Fig. 8.

The improved lignin penetration in ADESs can be attributed to its low viscosity and low-density. However, exceeding 20% water content resulted in diminished lignin removal.[53] It might be attributed to the excessive water reducing the DES density and disrupting the hydrogen bonding network, thereby limiting the contact between DESs and lignin. A similar observation was made by New *et al.* when investigating ChCl : U ADES.[18] Increasing the water content gradually decreased the viscosity and density, further enhancing penetration into lignocellulose and delignification. Nevertheless, when the water content exceeded 30% in ADES, the delignification effect was reversed. This is primarily due to the dilution of DES caused by a high amount of water, reducing the DESs concentration, and water molecules acting as a third component in the system, thereby reducing collisions between DESs and lignin.

4.4. Intensification techniques for lignin extraction

Except the reaction temperature and time, employing intensification technologies to assist lignin extraction in ADESs is a favorable option. Ultrasound, microwave, and ammonia fiber expansion blasting are commonly used for this purpose.[68]

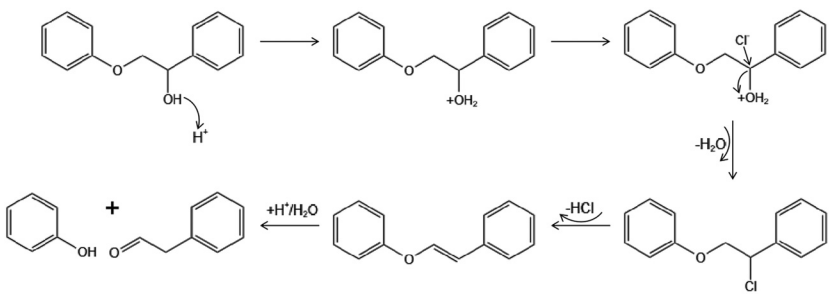


Fig. 7 The mechanism of the breaking of β-O-4 bonds by a Cl^- ion in PPE.

 This journal is © The Royal Society of Chemistry 2023

Published on 16 November 2023. Downloaded by Westlake University on 11/3/2025 9:33:15 AM.

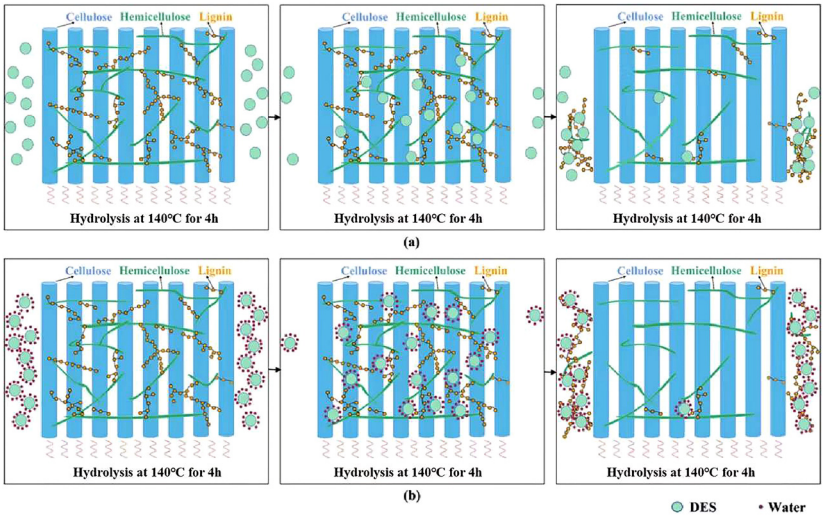


Fig. 8 Mechanism of delignification in (a) ChCl : LA DES and (b) ChCl : LA ADES (reproduced from ref. 53. Copyright (2022) Springer Heidelberg).

Ultrasound utilizes mechanical and cavitation effects to disrupt the lignocellulose structure and increase the contact area between ADESs and lignin. In a study by Ong *et al.*, ultrasound-assisted separation of lignin using ChCl : U ADES was employed.[65] Fig. 9 illustrates the mechanism of ADES action with ultrasound assistance, where low-viscosity ADES promotes the formation and growth of microbubbles. As the microbubbles reach a critical size, cavitation occurs, generating extreme pressure and excitation waves. These waves propagate toward the lignocellulosic surface, resulting in cell wall perforation and improved penetration of ADES into the lignin LCC bonds.

Microwave and ammonia fiber expansion blasting methods serve a similar purpose by physically disrupting the lignocellulose structure and enhancing the reaction surface area for ADESs treatment. Ultrafast microwave-assisted separation and extraction of lignin from corn stover using ChCl : LA DESs was reported by Chen *et al.*[69] Lignin was treated under microwave irradiation at a power of 800 W for 45 s, and the temperature was 152 °C at the end of 45 s. The lignin removal rate reached 79.60%, and the purity of lignin recovered from the treated solution was as high as 85–87%, with the cellulose retained and the digestibility increased by 2–5 folds. Liu *et al.* used microwave-assisted carboxylic acid-based DESs to pretreat corncobs.[70] Among them, ChCl : FA DESs were the most effective after microwave heating treatment at power 800 W for 3.5 min, with a lignin removal of 77.4%, a recovery yield of 64.7%, and a purity of 90.6%. It was found that the obtained D-lignin had an average particle size (136–141 nm), low molecular weight, high dispersibility, and exhibited good thermal stability. The structural characterization of D-lignin revealed that the S and G units were mainly present in D-lignin, and the phenolic OH content was significantly increased, which was favorable for the subsequent conversion and application of D-lignin.

Hu *et al.* developed a method for the rapid separation of wheat straw components using ammonia fiber expansion blasting-assisted ChCl : LA DES.[68] The one-pot pretreatment of ammonia fiber expansion blasting-assisted DES at 120 °C for 90 min resulted in an 88% removal rate of lignin and a 98%

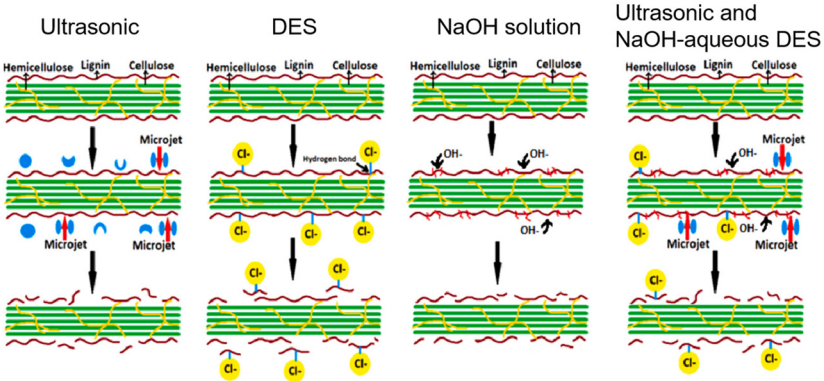


Fig. 9 The mechanism of action of ultrasound-assisted ADESs in lignin extraction (reproduced from ref. 65. Copyright (2021) Elsevier).

Published on 16 November 2023. Downloaded by Westlake University on 11/3/2025 9:33:15 AM.

enzymatic efficiency. The excellent treatment effect may be attributed to the opening of the tightly packed structure of cellulose, the disintegration of crystalline cellulose, and the partial disruption of hemicellulose by ammonia fiber expansion blasting pretreatment. Additionally, the extracted D-lignin has a low molecular weight and contains three structural units: H, G, and S.

5. Tools for the interaction of ADESs with lignin

Several tools and methods can be used to analyze and study lignin when investigating the separation and extraction of ADESs. Density functional theory (DFT) is a quantum chemical computational method that can be used to simulate and understand the structure, energy, and properties of molecules and materials. A commonly used tool is NMR technology, which can be used to study the interaction between solvents and lignin molecules. Also, FT-IR can be used to analyze solvent–lignin interactions. These tools and methods can help researchers to better understand the chemical reactions and interaction mechanisms between low-eutectic solvents and lignin. In this section, the tools and methods used to analyze and study the depolymerization mechanism of lignin, such as DFT, K–T parameters, NMR, FT-IR, and other tools and characterization methods, are introduced (Fig. 10).

5.1 DFT calculation

The DFT approach enables the calculation of electron density in a molecule, offering computational access to the interaction between lignin and DESs.[71] Lopes *et al.* conducted a study using DFT calculations (Fig. 11) to analyze the cleavage mechanism of PPE, a lignin model compound, in *p*TSA : ChCl and *p*TSA : ChBr ADES.[72] Their focus was on hydroxyl protonation, α-C dehydration, and β-carbon deprotonation, which involved the formation of enol-ether intermediates (double bonds between α and β carbons). Fig. 11 illustrates that the presence of Cl^- and Br^- favors the dehydration of PPE. During dehydration, the halide anion binds to α-C, stabilizing both the transition state structure and the dehydrated intermediate species. Comparing the reaction with and without halide anions, the transition state structure without halides is more than 12 kcal mol^{-1} higher, while the final state of dehydration is unstable (50 kcal mol^{-1}). However, the (1-chloro/bromo-2-phenoxy ethyl)benzene intermediates exhibit greater stability in the

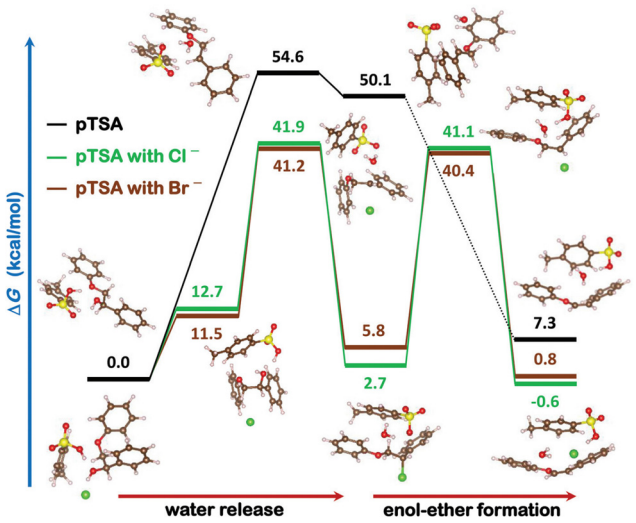


Fig. 11 The free energy curves of PPE cleavage in the presence of *p*TSA, *p*TSA with Cl^-, and *p*TSA with Br^-. Sphere color code: white for H, brown for C, red for O, yellow for S, and green for Cl^-. The sphere and rod models show the optimized structure of the reaction in the absence and presence of halides (only the chloride models are shown as they are similar to their bromide counterparts) (reproduced from ref. 72. Copyright (2020) The Royal Society of Chemistry).

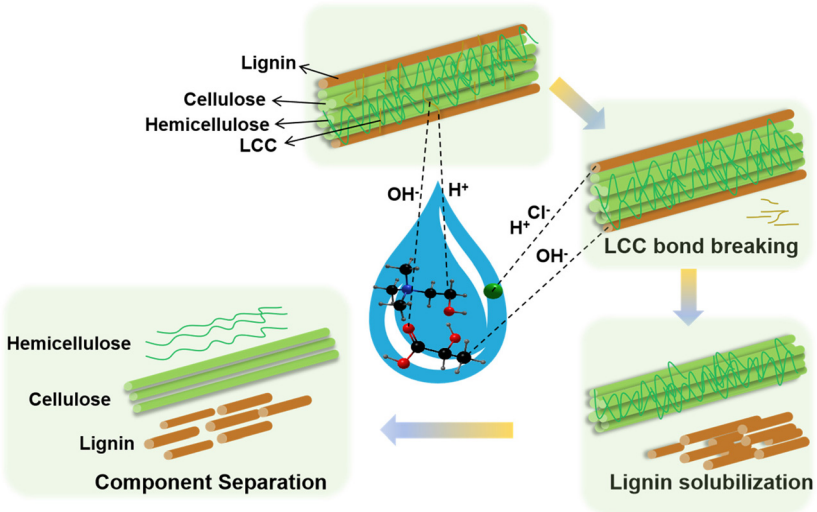


Fig. 10 The potential mechanisms for lignin solubilization by ADESs.

 This journal is © The Royal Society of Chemistry 2023

Published on 16 November 2023. Downloaded by Westlake University on 11/3/2025 9:33:15 AM.

presence of halides, at 2.7 kcal mol^{-1} and 5.8 kcal mol^{-1}, respectively.

When water is introduced into the DESs, the cleavage of the β-O-4 ether bond in PPE is inhibited, indicating that water addition protects the β-O-4 ether bond from breaking. It may be attributed to the prevention of hydroxyl protonation and subsequent dehydration steps in PPE in the presence of water. Water forms hydrogen bonds with Cl^- and Br^-, impeding their attack on the α-C site. Conversely, the halide counter ion effectively hinders the reaction of PPE with the PPE dimer resulting from dehydration, suggesting that the chloride counter ion might inhibit the aggregation of lignin.

Another study conducted by Xia *et al.* focused on the mechanism of lignin extraction using ChCl : Gly : $AlCl_3$ DESs and employed DFT to analyze the process, utilizing 1-(4-methoxyphenyl)-2-methoxyethanol as a lignin model compound (Fig. 12).[73] The findings revealed that Cl^- could form weak hydrogen bonds with –CH groups (CH-Cl, $\rho < 0.015$ au) and strong hydrogen bonds with –OH groups (OH-Cl, $0.028 < \rho < 0.036$ au). Interestingly, despite oxygen typically acting as a good HBA, the hydrogen bond strength of OH-Cl in lignin was found to be higher than that of the OH-O bond in lignin. It indicated that the Cl^- present in the $AlCl_3$ component of the DESs could compete with lignin to disrupt the hydrogen bonding network within lignocellulose, with a particular emphasis on breaking the weak ether bonds (C–O) present in lignin. The reaction pathway observed in this study was similar to that of acidic ADESs. Initially, the acidolysis of the DESs occurs, leading to the formation of positive protons at the α-C position. This step resembled the acid-catalyzed reactions often observed in lignin fractionation. Through subsequent deprotonation and protonation processes, a resonance-stabilized positive charge was formed. This was followed by the cleavage of the ether bond, resulting in the formation of a ketone. Finally, the ether bond was cleaved at the β-C position after undergoing hydration and deprotonation, leading to the formation of ketone and phenol groups.

In addition to DFT, other theoretical simulations, such as the conductor screening model for real solvents (COSMO-RS) model, quantum chemistry (QC) calculations, and MD simulations, can likewise be described, at the atomic level, the precise structures, energies, and interactions that can explain the mechanisms of fractionation, solvation, depolymerization, and polymerization of lignin in ADESs system.

5.2. Molecular dynamics

Molecular dynamics (MD) simulations can be used to study the mechanism of lignin depolymerization and modification by simulating the structure of the lignin molecule and the interactions between different functional groups, structural dynamics, solvent effects, and reaction simulations.

Radhakrishnan *et al.* used MD to study the molecular interactions during lignin solubilization in different (1-ethyl-3-methylimidazole acetate) (EimimOAc)-water mixtures.[74] The analysis showed that in pure water, the lignin formed compact aggregates. However, in 50%, 80%, and 100% EimimOAc-water mixtures, accompanied by extensive dynamic motions, lignin undergoes depolymerization *versus* condensation. The distribution of $Emim^+$, OAc^-, and water over the polar and nonpolar sites of the lignin unit was investigated, and effective lignin solubilization was found in 50% and 80% EimimOAc-water systems. The distribution of OAc^- was stronger around the alkyl chains and alcohol hydroxyl groups of lignin. The $Emim^+$ tails significantly affected the benzene rings, whereas the $Emim^+$ head was distributed around the methoxy and hydroxyl groups. MD provides new tools and methods for exploring the mechanistic studies of lignin extraction.

Wu *et al.* employed the solvent transfer method to prepare LNPs using a DES synthesized from sulfanilic acid and U, with water serving as the anti-solvent.[75] They established a theoretical model of the solvent system and a model for lignin compounds using quantum chemistry and MD theory, and subsequently, they analyzed the internal mechanism of nanoparticle self-assembly. During the self-assembly process, the dominant van der Waals interactions occurred between the model lignin compounds and sulfanilic acid within the system. As solvation progressed, the breakage of the β-O-4 bond in lignin, facilitated by hydrogen bond diffusion invol-

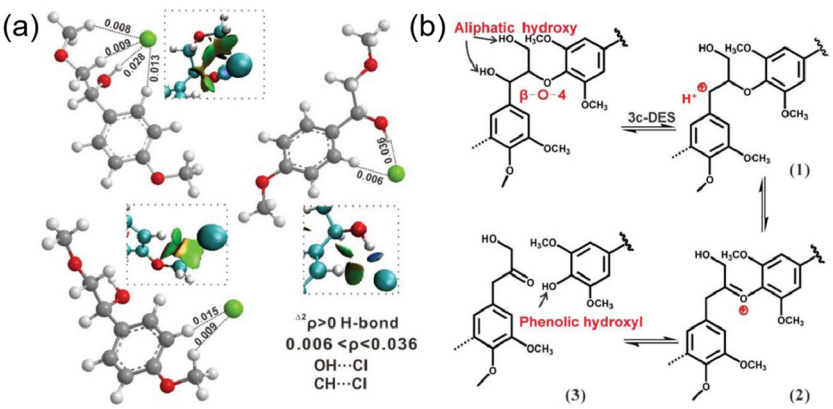


Fig. 12 (a) The calculated geometries of lignin model-Cl^- complexes at the M06-2X/6-31+G** level. (b) A schematic diagram showing the possible reaction pathway of the cleavage of the β-O-4 compound (reproduced from ref. 73. Copyright (2018) The Royal Society of Chemistry).

Published on 16 November 2023. Downloaded by Westlake University on 11/3/2025 9:33:15 AM.

ving sulfanilic acid and U within the DES, resulted in the release of more hydroxyl groups. The oxygen of the hydroxyl group in lignin formed intermolecular hydrogen bonds with the hydrogen of the amino group in DES, thereby enhancing the hydrophilicity of the lignin chain segments. This led to a remarkable amphiphilic nature of lignin in the mixed system. However, the introduction of water induced a proclivity for the polar portion of the model lignin compounds to associate with water. Consequently, the increased hydrophobicity of the system, the reduced total free energy of the surface in the solvent, and the heightened degree of entropic aggregation collectively enabled the hydrophobic backbone of the model lignin compounds to generate π–π stacking and assemble into spherical nanoparticles.

Hence, MD simulations play a pivotal role in elucidating the mechanism of lignin depolymerization by low eutectic solvents. They offer crucial atomic-level insights into the dissolution process, thereby contributing to a deeper comprehension of the underlying mechanisms. Furthermore, these simulations provide invaluable theoretical underpinnings for the enhancement and optimization of biomass conversion and lignin extraction processes.

5.3. Kamlet–Taft solvatochromic parameter

Kamlet–Taft (K–T) parameters, namely α, β, and π^*, are frequently used to characterize the chemical properties of DESs.[76]

The α parameter represents the hydrogen bond donating capability or acidity of the DESs. Higher α values indicate a higher ability to provide hydrogen protons, which can facilitate the protonation of functional groups like α-OH in lignin. This protonation plays a crucial role in catalyzing the cleavage of ether and ester linkages present in lignocellulose, thus aiding in lignin removal and extraction.

The β parameter, on the other hand, indicates the hydrogen bond accepting or alkalinity of the DESs. DESs with higher β values exhibit increased hydrogen bond receptivity, which can enhance their interaction with lignin molecules during the extraction processes.

The π^* parameter reflects the polarity or polarization of the solvent. It provides information about the solvent's ability to solvate various functional groups and compounds. The polarity of the DES affects its solubility and interaction with lignin during extraction.

5.4. FT-IR technology

FT-IR has a variety of applications in ADESs extraction lignin research, including lignin structure characterization, functional groups of lignin, and identification of intermediates, thus helping researchers gain insight into the chemical reactions and mechanisms of the lignin extraction process.

The FTIR spectra of D-lignin showed multiple absorption bands, indicating a high degree of inhomogeneity in its chemical structure. Tian *et al.* observed mainly lignin characteristic peaks such as –OH stretching (3425 cm^{-1}), –CH stretching (2930 and 2850 cm^{-1}), and aromatic backbone vibrations (1607 and 1514 cm^{-1}) in D-lignin.[77] Yang *et al.* observed signal peaks of stretching vibrations of the unconjugated carbon group in the carboxyl group near 1706 cm^{-1} as well as signals of asymmetric C–H deformations associated with the benzene ring in lignin at 1459 cm^{-1}.[78] Chen *et al.* found the lilac-based C–O vibration at 1240 cm^{-1}.[79] Yang *et al.* detected a signal representing the C–H deformation of lilac-based at 1126 cm^{-1}.[80]

FT-IR is a very useful tool for structural characterization and analysis of lignin. FT-IR is often used in combination with other analytical techniques, such as NMR and mass spectrometry, to obtain more comprehensive information.

5.5. NMR technology

The NMR technique has various applications in low eutectic solvent extraction of lignin, including ^{13}C-NMR, ^{1}H-NMR, and 2D HSQC NMR. Among them, ^{13}C and ^{1}H-NMR are often applied to the characterization of D-lignin, which can be used to analyze the carbon and hydrogen spectral profiles of lignin samples to determine the chemical environments and chemical shifts of the individual carbons and hydrogens in lignin, which can help to further analyze the structure of the lignin molecule and speculate on the mechanism of lignin depolymerization. Brandt *et al.*, employing the ^{1}H–^{13}C HSQC NMR technique, observed the complete disappearance of β-O-4 linkages, following a 12 h pretreatment of *Miscanthus giganteus* with acidic ionic liquids.[81] Similarly, Soares *et al.*, utilizing ^{1}H NMR, detected β-O-4 bond cleavage in acidic DESs during the delignification process, leading to the abundant formation of phenolic hydroxyl groups.[82] Chen *et al.* analyzed the mechanism of lignin extraction by ADESs using 2D HSQC NMR.[83] ChCl : EG ADES (10 wt% and 1.0 wt% H_2SO_4) was used to extract lignin at 110 °C with a β-O-4 linkage abundance of 15.0%, which was lower than the 20 wt% water content (24.4%), but when the temperature was increased to 130 °C, the lignin β-O-4 bond abundance decreased to zero. This is because the acid-catalyzed cleavage of the β-O-4 bond is triggered by the protonation at the Cα position, followed by dehydration and further decomposition of the ether bond. The higher temperature of 130 °C enhanced the reactivity of protons in the DES, ultimately leading to the complete cleavage of β-O-4 bonds within the lignin structure. Similarly, an increase in the aromatic region was found due to the breaking of the β-O-4 bond condensation reactions at the 2-, 5-, and 6-positions in the S and G units, thus increasing the S/G value.

Cheng *et al.* analyzed the aliphatic, phenolic, and carboxylic OH changes of lignin using ^{31}P NMR.[84] When the temperature was lower than 110 °C, the OH was almost unchanged. However, when the pretreatment temperature was further increased from 110 °C to 140 °C, the aliphatic OH content decreased significantly from 2.59 (110 °C) to 1.61 (120 °C) and 1.04 mmol g^{-1} (140 °C), whereas the phenolic OH increased from 1.19 (110 °C) to 1.49 (120 °C) and 1.69 mmol g^{-1} (140 °C), and the C5-substituted OH increased from 0.32 (110 °C) to 0.55 (120 °C) and 0.65 mmol g^{-1} (140 °C). This suggests a partial breakage of the β-O-4 structure and an

 This journal is © The Royal Society of Chemistry 2023

Published on 16 November 2023. Downloaded by Westlake University on 11/3/2025 9:33:15 AM.

increase in the relative content of phenolic OH compared to aliphatic OH, as well as more polymerization reactions under harsher pretreatment conditions, a result that is also in line with the findings of ^{1}H NMR.

Taken together, NMR techniques in the extraction of lignin by ADESs not only help to analyze the structure and composition of lignin but can also be used for the monitoring and optimization of the extraction process as well as the quality control of the extracted substances. This information is of great significance for the application and research of lignin.

6. Characterization of lignin extracted by ADESs

Using ADESs to break the hydrogen bonds with LCC could achieve ADES-regenerated lignin (referred to as D-lignin in this work).[85,86] Better understanding the characteristics of D-lignin can prompt its potential for valorization, including purity, molecular structure, chemical groups, molecular weight, and other properties.[87] D-lignin exhibits a similar chemical structure to alkali lignin and enzymatically dissolved lignin, with its benzene ring skeleton largely undamaged, thereby retaining a more intact basic structural unit.[16] However, the use of ADESs at high temperatures leads to the significant cleavage of β-O-4 bonds in lignin, with the S-unit-connected β-O-4 bonds particularly prone to breakage compared to the G-unit-connected β-O-4 bonds.[62] It is important to note that only a small fraction of C–C and C–O bonds is broken in comparison to β-O-4 bonds, which aligns with the theoretical bond energies of these bonds. Additionally, D-lignin undergoes minimal condensation reactions, resulting in a relatively low relative molecular mass (ranging from 490 to 2600) and a uniform distribution (polydispersity index <2).[43]

ADESs can effectively influence the particle size of D-lignin, which readily reaches micrometer and nanometer scales. Additionally, D-lignin exhibits a higher concentration of phenolic hydroxyl groups compared to the lignin obtained by other techniques, resulting in its exceptional activity.[78] These advantageous characteristics position D-lignin as a versatile material for generating lignin-based functional products and preparing aromatic compounds.

To evaluate these properties effectively, advanced analytical techniques such as FT-IR,[88] two-dimensional heteronuclear singular quantum correlation nuclear magnetic resonance (2D HSQC NMR),[89] gel permeation chromatography (GPC),[90] thermogravimetric analysis (TGA),[91] and GC/MS[92] can be employed. Leveraging the distinct attributes of D-lignin, its potential value can be explored extensively. This section will delve into the M_w, thermal stability, and structural feature of D-lignin.

6.1. Molecular weight

Molecular weight (M_w) and polydispersity index (PDI) are key characteristics that significantly impact the properties of lignin.[93] The size of the lignin molecules is correlated with their thermal stability and physical attributes. Compared to cellulose, which has a smaller molecular weight and is less thermally stable, lignin possesses a larger molecular weight, enhanced thermal stability, and a more robust structure.[94] The utilization of ADESs in lignin treatment leads to depolymerization, resulting in a decrease in lignin's M_w. However, when the treatment conditions become more severe, there is a possibility of repolymerization occurring, leading to an increase in M_w. The alteration in lignin's M_w is attributed to the dynamic interaction between the processes of depolymerization and polymerization, representing a competitive balance between the two. During ChCl : LA DES extraction, the M_w of *eucalyptus* lignin decreased from 6359 g mol^{-1} to 4817 g mol^{-1}. Subsequently, a slight further reduction in M_w to 4647 g mol^{-1} was observed with the addition of 20% water. It indicated that the presence of water in DESs not only caused additional lignin degradation but also potentially hindered lignin polymerization. The PDI of the lignin samples were 2.78, 3.77, and 2.99, respectively.[53] Similarly, the M_w of D-lignin decreased after ChCl : EG ADES treatment (10% water), and the treated D-lignin exhibited a reduced M_w of 3932.5 g mol^{-1} compared to natural lignin (9544.5 g mol^{-1}), accompanied by a contraction in PDI.[95]

In addition to water, the type of HBA and HBD in DESs, as well as most ternary DESs formulations, can also reduce the M_w of lignin.[96,97] Increasing the temperature and treatment duration have a similar effect on reducing the M_w of D-lignin.[52,98] Another crucial factor affecting the M_w of D-lignin is the solid loading, which tends to increase with higher solid loading due to the limited mass transfer effect of ADES during the treatment. This limited mass transfer effect, along with the minimal cleavage of ether bonds, results in a weak depolymerization effect on lignin.[99]

6.2 Thermal stability

D-lignin tends to decompose at higher temperatures due to the partial cleavage of weak aromatic ether and ester bonds during the pretreatment of ADESs extraction, leaving behind more stable bonds within D-lignin.[99] The thermal stability of D-Lignin is crucial for its use in lignin-based products.[83] Apart from the cleavage of ether bonds, secondary reactions can occur during ADESs pretreatment, leading to condensation and re-polymerization reactions in lignin.[100] TGA curves showed that D-lignin extracted using ChCl : LA ADES at 700 °C experienced a weight loss of 40.7 wt%, while alkali lignin extraction resulted in a weight loss of 43.3 wt%. It indicated that D-lignin exhibited greater thermal stability compared to alkali lignin.[48] The cleavage of intermolecular bonds between lignin during ADESs treatment results in smaller fragments of D-lignin, as observed by scanning electron microscopy (SEM) images. Due to these excellent properties, the D-lignin extracted through ADESs treatment retains the lignin backbone, possesses a narrow molecular weight distribution at low M_w, and demonstrates enhanced thermal stability. These characteristics make D-lignin well-suited for various applications involving functionalization.[48]

Published on 16 November 2023. Downloaded by Westlake University on 11/3/2025 9:33:15 AM.

6.3. Structural features

Lignin contains various functional groups such as aryl, alcohol hydroxyl, phenol hydroxyl, carboxyl, aldehyde, and carbonyl groups. These functional groups, along with lignin structural monomers, are primarily connected through β-O-4, β–β, β-5, β-1, 5–5, and 4-O-5 bonds.[101] When lignin molecules isolated in ADESs underwent various reactions, they formed different ratios and types of functional groups and bonding connections. Some of the main functional groups observed in D-lignin are –OH and $-OCH_3$ groups. To analyze the chemical structure and its changes, 2D-HSQC NMR is commonly used, which provides information about hydrogen (H), carbon (C), and phosphorus (P) interactions. D-lignin exhibits characteristic peaks related to lignin in both the aromatic region ($\delta C/\delta H$ 100.0–150.0/9.0–5.5) and the side-chain region ($\delta C/\delta H$ 50.0–90.0/2.50–6.0).[68] In a study by Cui *et al.*, a comparison between D-lignin and milled wood lignin was conducted.[53] The partitioning of connecting bonds, such as β-O-4, β–β, and β-5, in the lignin units was relatively weakened or disappeared in D-lignin. This is because the ether bond (C–C bond) in the lignin-hemicellulose linkage is broken during the ADES treatment (Fig. 13a). However, in D-lignin, the β-O-4 bond remains the most dominant connecting bond between lignin units. The aromatic region of D-lignin showed an increase in the number of subunits of condensed lignin up to 50.7%. On the other hand, the number of subunits of D-lignin without water was 63.8%, indicating that water prevents lignin from undergoing polymerization. Consequently, less condensed and more stable lignin could be obtained in ADESs.

It has been demonstrated that the presence of water in ADESs is associated with the β-O-4 bond cleavage. Cheng *et al.* conducted a comparison between enzymatically cleaved lignin and D-lignin.[56] The β-O-4 bond content in enzymatically extracted lignin was found to be 59/100Ar, whereas in D-lignin extracted using ChCl : BDO ADES, it significantly decreased to 5/100Ar. The content of β-O-4 bonds increased as the water content in ADES increased, reaching 14/100Ar (20%), 18/100Ar (40%), and 27/100Ar (60%). The introduction of water leads to a decrease in the catalytic activity of acidic protons due to the solvation of water molecules. Apart from water content, the solid–liquid ratio also impacts the β-O-4 bond content. The solid–liquid ratio primarily affected mass transfer during ADESs treatment.[78] Lower solids loading enables better contact between ADESs and lignin, enhancing lignin separation efficiency. Conversely, higher solids loading impairs ADESs permeation ability and reduces lignin depolymerization.[99] The strong $-OCH_3$ signals in 2D-HSQC NMR spectrum indicate the presence of a considerable number of $-OCH_3$ groups in D-lignin, making it a suitable choice for antioxidant applications.[95] Lignin predominantly comprises three aromatic units-syringyl (S), guaiacyl (G), and *p*-hydroxyphenyl (H) units.[102] When the reaction temperature exceeded 140 °C, the S/G ratio decreased, indicating the concurrent degradation reactions of the S and G units.[27] Additionally, signals from *p*-coumarate (*p*-CA) and ferulic acid can be observed in the aromatic region, representing the ether and ester bonds that link lignin to hemicellulose and cellulose, forming the LCC. As seen from Fig. 13(b), the decrease in ferulic acid content in the 2D-HSQC NMR spectrum hinted the disruption of LCC bonds.[56]

The phenolic hydroxyl group is an essential component of the lignin structure and has a significant influence on the physicochemical properties of lignin.[103] When lignin was extracted using ADESs, the content of phenolic –OH groups could be increased by breaking the β-O-4 bond. The quantification of different hydroxyl functional groups, including aliphatic, phenolic, and carboxylic OH, is typically performed using ^{31}P NMR spectroscopy. Among these different hydroxyl groups, aliphatic OH is the most abundant, comprising approximately 60%–70% of the total hydroxyl groups. The content of phenolic –OH groups in enzymatically digested lignin was reported to be 23.6%. However, during the ADES extraction of D-lignin, the content of phenolic –OH groups

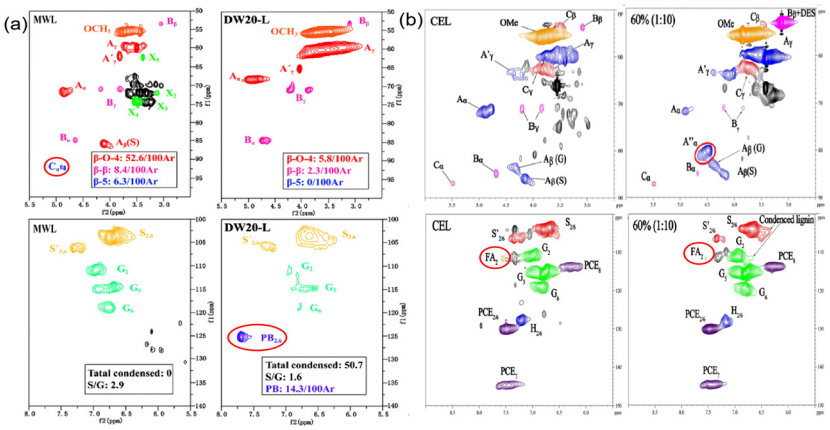


Fig. 13 The 2D HSQC NMR spectra of D-lignin. (a) ChCl : LA ADES (20 vol% H_2O) extracted lignin compared with milled wood lignin. (b) ChCl : BOD ADES (60 wt% H_2O) compared with enzymatically extracted lignin (reproduced from ref. 53 and 56. Copyright (2022, 2023) Springer Heidelberg and The Royal Society of Chemistry).

 This journal is © The Royal Society of Chemistry 2023

Published on 16 November 2023. Downloaded by Westlake University on 11/3/2025 9:33:15 AM.

decreased slightly with increasing water content, from 30.5% (10% H_2O) to 28.2% (60% H_2O).[122] This suggested that high water content impeded the ability of ADESs to break the β-O-4 bond.[56] The term "C5-substituted –OH" refers to hydroxyl groups resulting from the condensation of the G group with other 5-substituted unit types, such as 5–5 and β-5 linkages. The content of C5-substituted OH decreased from 8.3% to 6.6% as the water content in ADESs increased from 10% to 60%. Diol DESs were found to significantly hinder the lignin condensation reaction during pretreatment, leading to a decrease in C5-substituted OH content.[56]

7. Applications for valorization of lignin extracted using ADESs

Although natural lignin products are expected to be used in the fabrication of various polymer-based functional composites, their high value-added utilization remains a challenge due to their structural complexity and low localization activity. The efficient separation, extraction, modification, and depolymerization of lignin using DESs is not only conducive to the improvement of the enzymatic saccharification efficiency of cellulose but also facilitates the preparation of D-lignin with high yield, high purity, and uniform molecular mass distribution, which ensures the basic high-value utilization of D-lignin. D-lignin also has a wide range of applications in the fields of catalysis, materials, medicine, and environment. Therefore, the high-value utilization of D-lignin has far-reaching implications for DESs in biomass biorefineries. In this section, the applications of D-lignin in the preparation of lignin nanoparticles, UV-resistant materials, and drug delivery will be discussed. In this section, the applications of D-lignin in the preparation of lignin nanoparticles, UV-resistant materials, and drug delivery will be discussed.

7.1. Lignin nanoparticles (LNPs)

Lignin nanoparticles (LNPs), also known as lignin nanospheres, are a promising class of eco-friendly products with vast potential across various fields including drug delivery,[104] food industry,[105] environmental applications,[106] and biomedicine.[107] Table 3 summarizes the properties of LNPs derived from different types of ADESs. Guo *et al.* employed two acidic DESs (ChCl : FA and ChCl : LA) as well as two basic DESs (ChCl : MEA and Gly : K_2CO_3) to extract D-lignin from bamboo waste and successfully created LNPs through dialysis precipitation self-assembly.[108] The findings indicated that DESs were capable of breaking down the LCC and lignin-ether bonds, facilitating the separation of the extracted lignin. The chemical structures and compositions of the four D-lignins were all modified, resulting in variations in van der Waals interactions, π–π stacking, and hydrophobic interactions, which impacted the properties of the resulting LNPs. Notably, the diameter, M_w, hydrophobicity, and thermal stability of LNPs derived from D-lignin were compared. LNPs derived from D-lignin extracted using alkaline DESs exhibited smaller sizes, with average sphere diameters ranging from 50 nm to 100 nm. The diameter of LNPs was predominantly associated with the M_w of lignin, and among the four D-lignins, the ones extracted using ChCl : MEA and Gly : K_2CO_3 exhibited larger M_w. The larger size of LNPs prepared using ChCl : MEA DESs could be attributed to the impact of the amine group on the electron density of the aromatic ring, which weakened π–π stacking. On the other hand, acidic DESs subjected to harsh conditions typically resulted in lower M_w D-lignin, which weakened van der Waals forces and π–π stacking. However, the condensation between lignin fragments enhanced hydrophobic interactions, leading to the higher thermal stability of the prepared LNPs.

Luo *et al.* employed the dialysis precipitation self-assembly technique to create LNPs from D-lignin, yielding sub-100 nm functionalized LNPs.[109] The resulting LNPs exhibited stability

Table 3 The preparation and properties of LNPs obtained by different DESs

DESs composition	Biomass	Preparation conditions	Mechanical method	Particle size (nm)	Yield (%)	Ref.
ChCl : LA	Wheat straw	120 °C, 12 h, 1 : 10	Solvent exchange precipitation self-assembly	50–150	85.9	151
ChCl : OA	Bamboo powders	90 °C, 3 h, 1 : 10	Ultrasonic self-assembly method	584	82.1	152
ChCl : LA	Corncob residue	150 °C, 18 h, 1 : 10	Solvent exchange precipitation self-assembly	48.66	85.6	153
ChCl : LA	Rice straw	Microwave 680 W, 3 min, 1 : 30	Solvent exchange precipitation self-assembly	48–95	11	154
ChCl : MA	Bamboo	110 °C, 10 min, 1 : 15	Dialysis precipitation self-assembly	20–180	89.67	155
B : LA	Dealkalized lignin	25 °C, 12 h, 1 : 0.2	Solvent exchange precipitation self-assembly	57.16	74.8	156
ChCl : MA	*Eucommia ulmoides* Oliver wood	130 °C, 3 h, 1 : 10	Solvent exchange precipitation self-assembly	60–110	—	157
BTMAC/FA	Radiata pine	150 °C, 2 h, 1 : 20	Solvent exchange precipitation self-assembly	95 nm	80.6	158
ChCl : TsOH : EG	Walnut shells	90 °C, 2 h, 1 : 15	Solvent exchange precipitation self-assembly	157.71	91.55	159
ChCl : EG	*Musa textilis*	110 °C, 30 min, 1 : 10	Solvent exchange precipitation self-assembly	30–50	56.23	160

Published on 16 November 2023. Downloaded by Westlake University on 11/3/2025 9:33:15 AM.

over 30 days. Incorporating these LNPs into polyvinyl alcohol (PVA) led to nanocomposite films showcasing robust mechanical properties (82.5 MPa tensile strength, 103.3% strain at break), complete ultraviolet (UV) protection, notable hydrophobicity (117.0° static contact angle), and heightened thermal stability (40 °C increase in max heat loss temp). Zhang *et al.* utilized a solvent exchange precipitation self-assembly approach to produce LNPs from D-lignin obtained through ChCl : LA DES, achieving remarkable yield (85.6%) and purity (97.8%) from corn kernel residue.[110] The resulting LNPs displayed uniform, small size with clear shapes, and high stability. Their hydrodynamic diameter was 48.66 nm, PDI below 0.2, and zeta potential exceeding 42.2 mV. Xu *et al.* pretreated poplar wood biomass using microwave-assisted ternary DESs, and then crafted LNPs from isolated lignin through solvent-exchange precipitation self-assembly.[111] Two DESs, ChCl : Gly : $AlCl_3$ and ChCl : EG : $AlCl_3$, produced controllable sub-100 nm LNPs. ChCl : EG DES yielded LNPs primarily sized 40–80 nm, while ChCl : Gly : $AlCl_3$ DES resulted in predominantly 30–70 nm LNPs, linked to D-lignin's M_w and structural traits. Modulating LNPs size was achievable by altering the DES ratios and reaction conditions.

All in all, LNPs exhibit a diverse array of applications, encompassing metal carriers, drug carriers, drug retardants, ultraviolet absorbers, and hydrogel matrices. The broad spectrum of applications for LNPs creates numerous opportunities for the high-value utilization of D-lignin.

7.2. Lignin-based materials with UV resistance

The various wavelengths of ultraviolet radiation are commonly classified into three categories: UV (320–340 nm), UVB (280–320 nm), and UVC (100–280 nm).[112] Among these, UVC radiation has strong penetration capabilities, with over 98% being able to permeate the ozone layer and clouds, reaching the Earth's surface. This poses significant risks to human health, as well as impacting the growth and development of plants and animals, while also accelerating the aging and degradation of commonly used materials.[113]

The presence of a conjugated system, hydrogen bonding, and specific structural features in lignin, such as benzene rings, methoxy groups, and phenolic hydroxyl groups, contribute to its effective UV shielding properties.[114] These structural elements and functional groups play a protective role against different wavelengths of UV light, enabling lignin to provide comprehensive UV protection. Additionally, the phenolic hydroxyl groups in lignin possess antioxidant properties, effectively scavenging free radicals.[115] Consequently, lignin exhibits promising potential as a UV protection material. In a study conducted by Yang *et al.*, a UV-shielding membrane was synthesized using D-lignin extracted from ChCl : LA DES with the addition of acetic acid and chitosan.[87] The resulting UV-shielding film containing 7.0% D-lignin exhibited a UV transmittance of 79.1% and haze of 8.9%, while a UV-shielding film without lignin had a transmittance of 92.3% and no haze. These findings indicate that the inclusion of D-lignin did not adversely affect the imaging or color of the films, demonstrating its compatibility as a UV-shielding material. However, as the concentration of lignin increased, the UV shielding performance did not show significant improvement, but the haze increased noticeably. This observation could be attributed to the saturation of lignin solubility in the solution, leading to an inhomogeneous film structure that affects the practical application of the material.

Li *et al.* conducted a study where lignin/ZnO composites were prepared using a one-pot method, utilizing U : $ZnCl_2$ DES as both the solvent and raw material. These composites were then blended with waterborne polyurethane (WPU) to form films, allowing for the investigation of their UV resistance properties.[116] MD simulations revealed that the interaction between lignosulfonate and U : $ZnCl_2$ DES primarily involved electrostatic forces, while enzymatic hydrolysis lignin interacted with U : $ZnCl_2$ DES through van der Waals interactions. The UV transmittance of WPU membranes incorporating lignin was significantly lower than that of pure WPU membranes (which exhibited a UV transmittance of 89.8%) within the UV wavelength range of 250–400 nm. This finding indicated that the addition of lignin improved the UV resistance of the WPU membranes. Moreover, the lignin-based WPU membranes displayed favorable mechanical properties, including an elongation-at-break value of 25.53% and a tensile strength of 1422 kPa, while maintaining good light transmittance.

The integration of D-lignin into sunscreens showcases a promising avenue of research, primarily attributed to its enduring ability to block harmful UV radiation. Qiu *et al.*'s research team embarked on constructing photostimulated UV-absorption-enhancing sunscreen actives (AL-SP) by covalently merging alkali lignin with active spiro-pyran (SP-Br).[136] This strategic combination substantially augmented the absorption rate of AL within the UVA range. By substituting certain phenolic hydroxyl groups in alkali lignin with SP-Br, the issue of aggregation was mitigated. The inherent antioxidant properties and biocompatibility of lignin endowed AL-SP with favorable cyclic stability and remarkable *ex vivo* biocompatibility. Consequently, lignin-derived sunscreens hold significant potential for practical applications (Fig. 14).

7.3 Drug delivery

With the progress of nanoscience and technology, lignin has emerged as a promising avenue for high-end applications due to its renewable, biocompatible, and non-toxic nature.[117] Notably, LNPs have proven to be highly effective carriers and delivery vehicles for active pharmaceutical ingredients, revolutionizing drug delivery, and release processes.[118] LNPs have also demonstrated significant potential in the agrochemical sector by showcasing strong stability, compatibility and the ability to encapsulate insecticides and fungicides without exacerbating environmental pollution. Shen *et al.* expanded the realms of biorefinery utilizing D-lignin extracted from DESs for the production of microencapsulated pesticides.[119] Fig. 15 illustrates the preparation of LNPs from DES-extracted lignin and their subsequent use in packaging avermectin pesticide. The addition of $FeCl_3$ to ChCl : LA DES resulted in a

 This journal is © The Royal Society of Chemistry 2023

Published on 16 November 2023. Downloaded by Westlake University on 11/3/2025 9:33:15 AM.

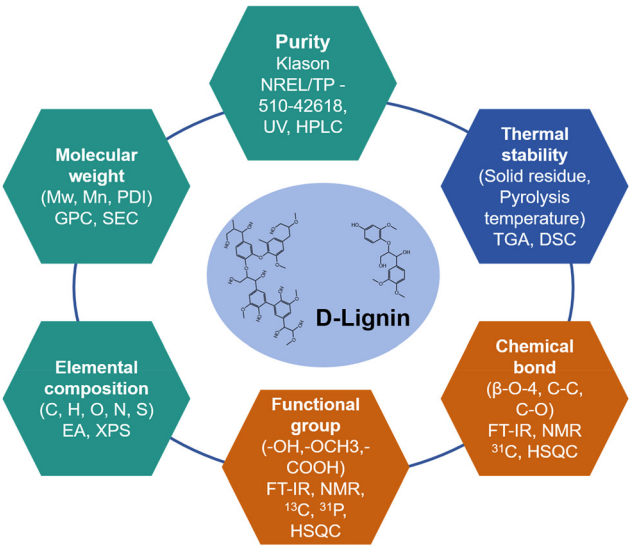


Fig. 14 The characterization of D-lignin structural properties analyzed by different methods.

70.8% increase in $FeCl_3$-mediated delignification of DES. This pretreatment process induced acid-catalyzed condensation and cleavage reactions of the lignin's β-O-4 bonds, yielding abundant condensed structures and phenol-OH groups that formed unique aromatic structures. Furthermore, the low M_w D-lignans with high phenol-OH content enhanced the photolysis resistance of LNPs-encapsulated avermectin (AVM@LNP) pesticides. The D-lignans exhibited robust π–π stacking, facilitating non-radiative migration and triggering photo-thermal conversion. Unlike unencapsulated AVM solutions, AVM@LNP showcased controlled release properties and temperature-responsive behavior, ensuring sustained pest control.

Jiang *et al.* developed lignin–xylan hybrid nanoparticles (LXNPs) from D-lignin through a straightforward solution self-assembly process and explored their potential as pesticide carriers for enzyme-mediated controlled release.[120] The LXNPs, synthesized from D-lignin obtained using ChCl : LA DES, exhibited particle sizes ranging from approximately 166–210 nm and demonstrated significant stability within the pH range of 4–10. The self-assembly mechanism of D-lignin into LXNPs led to the formation of a core–shell structure, as depicted in Fig. 16, with abundant hydroxyl groups on the surface. The binding of xylan fragments to D-lignin occurred through van der Waals forces and hydrophobic interactions between lignin phenylpropanes and the xylan molecular framework, resulting in the distinctive spherical core–shell architecture. The hydrophobic cores of the LXNPs readily encapsulated the biopesticide AVMs, achieving encapsulation rates of 57.9%–67.0%. Following 16 h of enzymatic digestion with xylanase, the release of AVM from these AVM-encapsulated LXNPs reached 44.8%–55.1%, in contrast to a mere 4.1% release from unencapsulated AVMs. This study introduced a nanospheric carrier for drug encapsulation and release, along with a promising utilization approach for D-lignin.

Liu *et al.*[121] synthesized D-lignin grafted poly(NVCL) (DES-lignin-*g*-PNVCL) through electron transfer-atom transfer radical polymerization (ARGET-ATPR), utilizing modified D-lignin and *N*-vinyl caprolactam (NVCL). Employing the ethanol/water anti-solvent method, they achieved the self-assembly of aspirin on D-lignin-g-PNVCL, resulting in lignin thermo-sensitive polymer nanoparticle-coated aspirin (aspirin@LTNP). The encapsulation rate of aspirin@LTNP

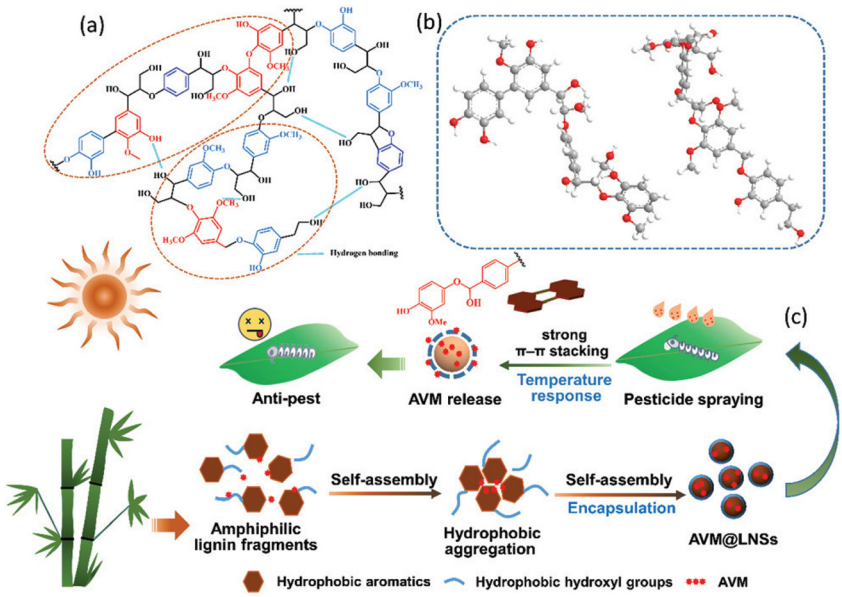


Fig. 15 The mechanism of LNPs regulating the controlled release of AVM insecticides through temperature-responsive properties. (a) The molecular chemical structure of lignin fragments; (b) three-dimensional structural diagram; (c) the process of LNPs from encapsulation to AVM release (reproduced from ref. 119. Copyright (2022) The Royal Society of Chemistry).

Published on 16 November 2023. Downloaded by Westlake University on 11/3/2025 9:33:15 AM.

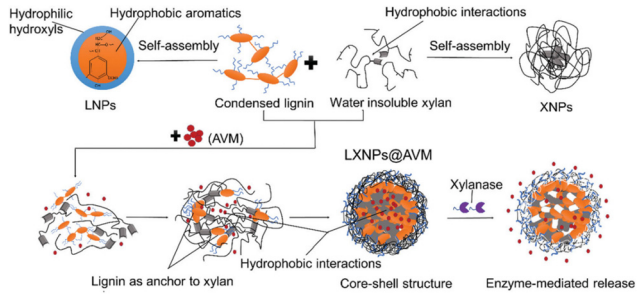


Fig. 16 A D-lignin self-assembly to form lignin–xylan hybrid nanospheres and its application toward avermectin encapsulation and xylanase-mediated release (reproduced from ref. 120. Copyright (2020) The Royal Society of Chemistry).

reached 88.87%. Notably, aspirin@LTNP exhibited a significant temperature response, with a cumulative release rate of 73.75% over 96 hours at the lowest critical solution temperature. Within acidic and alkaline solutions, distinct cumulative release rates were observed, with a rate of 63.21% ± 0.57% at pH 1.5 and 49.56% ± 0.48% at pH 7.4 over a period of 96 h.

8. Enhancement of the lignin extraction process using ADESs

The process of lignin extraction from ADESs consists mainly of dissolution and separation of lignin, recovery of DESs, and preparation of lignin products. Among them, the recovery of DESs and the preparation of lignin products are crucial for sustainable development. The effective recovery of DESs helps to reduce resource waste and environmental impacts, while diverse lignin product preparation promotes a sustainable bioeconomy and green chemical industry. Further research and improvements in these areas will lead to a wider application of lignin and provide strong support for sustainable development goals. Therefore, this section will discuss the recycling of DESs and the preparation and development of lignin products.

8.1. DESs recycling

DESs are called "green solvents" because of two significant features. Firstly, due to the low toxicity of HBDs and HBAs, DESs are less hazardous to human beings and the environment during their use. Secondly, the most important feature of DESs is that they can be recovered, recycled, and reused, which helps to reduce the waste of resources and environmental pollution, thus promoting the sustainable development of DESs in the biorefining industry.

Spinning or distillation is the most commonly used method for the separation and recovery of DESs.[32] Poy *et al.* used a ternary DES consisting of ChCl : *p*TSA : EG to treat rice straw for lignin removal.[122] After two cycles, this ternary DES achieved an average recovery of 85% w/w, but the lignin removal decreased from the initial 61.6% to 37.5%. The reason for the decrease in the performance of this DES may be related to the accumulation of impurities and the weakening of its hydrogen-bonding interaction ability. It is noteworthy that the saccharification rate of cellulose remained above 81% after the enzymatic digestion of cellulose with no obvious decreasing trend. Wang *et al.* investigated the disruption of the microstructure of poplar cell walls and the lignin solubilization process by ChCl : LA DES using a combination of microscopic and chemical analyses.[123] After four cycles, the lignin removal rate decreased slightly due to the weakening of acidic and hydrogen bonding interactions in the DES solution, but the recovery of lignin increased from 20.85% to 55.74%. This was attributed to less β-O-4 bond breakage in the removed lignin. The chemical structures of fresh and recovered DES were analyzed by ^{1}H and ^{13}C NMR. The NMR spectra of fresh and recycled DES showed similar chemical shifts of hydrogen and carbon atoms, but subtle intensity differences appeared due to the weakening of hydrogen bonding interactions in the recycled DES. Sun *et al.* pretreated *Pennisetum giganteum* biomass using ChCl : MEA DES for five cycles, and the recovery of the recovered DES was above 90% each time.[124] Analysis by FT-IR spectroscopy revealed that the hydroxyl peak wavelengths of both fresh and recycled DES were red-shifted compared to those of ChCl and MEA, suggesting that the hydrogen-bonding interactions of ester bonds occurred between them. Despite the reduced hydrogen bonding interactions, there was a small difference between the hydroxyl vibrational peaks between the fresh and recycled DES. However, it was also shown that the properties of the DES in each recycling cycle were completely similar, and no degradation products such as cellulose, hemicellulose, and lignin were detected.[125] Huang *et al.* pretreated sugarcane bagasse using ChCl : PEG : HSA DES.[126] After five cycles, the recovery of DES reached 95% and the removal of lignin remained above 60% in all cases. The $ZnCl_2$: LA DES used by Bai *et al.* showed essentially unchanged lignin removal from poplar sawdust after five cycles, from 95.2 to 91.2 wt%, and recovered lignin purity of 87.3%.[127]

Therefore, the recovery and efficient recycling of DESs can effectively reduce production costs, minimize waste emissions, and reduce dependence on limited resources. This sustainable chemical process is expected to provide important support for the green chemical industry and sustainable development in various fields, including biomass conversion, lignin extraction, and waste treatment. However, there is still room for improvement in the current recovery and recycling of DESs for delignification. By continuously improving the recovery and recycling methods of DESs, we can not only achieve a more environment friendly and economically viable production method but also help to promote the realization of a green economy and the sustainable use of resources.

8.2. Production of lignin-based products

The preparation of lignin-based products is another important area within the field of biorefining of ADESs. The structural

 This journal is © The Royal Society of Chemistry 2023

Published on 16 November 2023. Downloaded by Westlake University on 11/3/2025 9:33:15 AM.

diversity of lignin allows it to be further enhanced through chemical modification and functionalization to meet the needs of different fields. This diversity provides a wide range of opportunities for the valorization of lignin for applications including the production of high-value-added products such as aviation fuels, biomaterials, and platform compounds. Therefore, research in this area aims to improve the conversion efficiency of lignin to create high-value-added products that offer unlimited potential and possibilities for the biorefinery industry.

Diao *et al.* developed a catalyst based on Mo-doped Co_9S_8 nanoparticles anchored on an Al_2O_3 carrier ($Mo\text{-}Co_9S_8/Al_2O_3$) aimed at the selective conversion of lignin to aromatics.[128] This catalyst exhibited excellent hydrolytic deoxygenation activity, selectivity, and stability. In the hydrolytic deoxygenation of diphenyl ether, the catalyst achieved 99.8% conversion and 91.0% benzene yield after at least 10 consecutive reaction runs at a temperature of 265 °C. The structure of the $Mo\text{–}Co_9S_8/Al_2O_3$ catalyst forms Mo–S–Co type covalent bonds, which enhances its adsorption and activation of oxygenated substrates and efficiently realizes the cleavage of C–O bonds and the formation of C–O bonds. The C–O bond is cleaved, while avoiding the unnecessary hydrogenation of the benzene ring. When applied to the catalytic conversion of poplar lignin, the total carbon yield of the monomer was 14.8%, of which 9.1% were aromatic hydrocarbons and 5.6% were cycloalkanes. This study provides new approaches and insights for the development of efficient and stable catalysts for the conversion of lignin to aromatics. Sunthornvarabhas *et al.* used extracted lignin for the preparation of potential additional fabrics in masks, which successfully inhibited the growth of *Staphylococci* to the level of sterilization within 24 h.[129] The results of this study suggest that lignin has potential antimicrobial properties and offers new applications for the preparation of masks and other medical uses. Xi *et al.* prepared layered lignin-derived porous carbon materials by gas stripping and *in situ* templating techniques using $ZnCO_3$ activation.[130] The activation by gas release from $ZnCO_3$ gave it an ideal microstructure and a large number of mesopores. This layered porous structure of carbon with predominant mesopores can provide a mediated pathway for electrolyte ion diffusion and abundant active sites for lithium-ion storage. As a result, the anode of the carbon material has excellent capacity performance, with a high reversible capacity of 730 mA h cm^{-3} at a current density of 0.2 A g^{-1} and excellent cycling performance (retention of about 99% after 10 000 cycles). Research and development in this area is expected to drive innovation in the field of sustainable energy and materials and to make a significant contribution to green technology and environmental industries. These results provide new ideas and prospects for the high-value utilization of lignin and are expected to lead to important breakthroughs in the field of energy storage. Gebreyohannes *et al.*[131] developed a sustainable and eco-friendly lignin film by dissolving lignin into PA : U DES. The resulting lignin membrane exhibited a M_w cutoff of approximately 1.3 kg mol^{-1}, enabling the separation of molecules of similar sizes in both water and organic solvents. In another study, Akond *et al.*[132] utilized ChCl : FA DES for lignin extraction from bagasse and incorporated D-lignin into cement to enhance its overall performance. Xue *et al.*[133] designed ChCl : Gly : PEG DESs to efficiently dissolve lignin and synthesize lignin-containing rigid polyurethane foam.

By continuing to enhance the process of lignin extraction from ADESs, we can achieve more sustainable lignin extraction and high-value utilization and promote the sustainable development of biomass resources.

9. Sustainability considerations of lignin extracted by ADESs

The sustainability of ADESs in biomass refining processes relies on two key factors: the economic cost of ADESs and their environmental impact.[134] Conducting comprehensive assessments such as life cycle assessment (LCA) and techno-economic assessment (TEA) is essential to determine the sustainability of ADES in the lignin extraction.[135] Therefore, this section discusses the environmental friendliness and economic sustainability of ADESs in the field of biorefining.

9.1. Economic feasibility

The efficiency and direction of biomass biorefinery is determined by the production of fuels and chemicals from lignocellulose.[136] An ideal biorefinery process should be energy-efficient and cost-effective, while minimizing the consumption of active components in the raw material and yielding easily manageable products with low by-products.[137] Therefore, future researches on lignin extraction using ADES should focus on developing affordable, environment friendly, and recyclable ADESs solutions. Typically, the process of lignocellulose dissociation using ADESs involves three steps: (1) pretreatment for component separation, (2) solid–liquid separation, and (3) regeneration of dissolved components by introducing antisolvent solvents. The economic feasibility of D-lignin extracted using ADESs is mainly evidenced by the recovery of ADESs. Numerous excellent articles have reported some outstanding methods to recover DESs solvent. For instance, Dugoni *et al.* conducted four cycles of choline acetate (ChOAc) : levulinic acid (LevA) ADES, evaluating DES recovery through mass determination, and found an approximate 90% recovery per cycle.[138] Chourasia *et al.* employed vacuum evaporation of water for DES recovery.[42] The recoveries of pure DESs were found to be 86.7%, 93.7%, and 95.1% for DESs containing 20 w% aqueous solutions and 40 w% DESs, respectively. Increasing the water content was observed to enhance the DES recovery. It was demonstrated that even after four cycles, the DESs exhibited good activity, with a slight decrease observed in the fifth cycle. Various methods have been reported for DES recycling, including membrane processes, electrodialysis, and evaporation of liquids. These recovery

Published on 16 November 2023. Downloaded by Westlake University on 11/3/2025 9:33:15 AM.

methods have different implications for production cost and equipment requirements. Chen *et al.* conducted a study on the recovery of lignin-derived ChCl : *p*-coumaric acid (PCA) DESs using vacuum spinning.[139] The characterization of the recovered DESs using techniques such as FT-IR, TGA, and differential scanning calorimetry (DSC) revealed a significant reduction in hydrogen bonding compared to the original DESs. This suggests that the structure and properties of the DESs were altered during the recycling process. Recycling technologies for DESs are important for the economic sustainability of lignin extraction.

Furthermore, it is notable that derivatives of separated biomass can be utilized to formulate ADESs, enabling closed-loop biomass refining and improving the overall production efficiency. Efficient DESs recovery methods have the potential to significantly reduce the cost of biomass refining and contribute to the sustainable utilization of biomass resources.

9.2. Environment friendly aspects

DESs are favored due to their environment friendly nature, cost-effectiveness, design flexibility, and biodegradability. In the context of environmental sustainability, LCA is a comprehensive and systematic analytical approach that aids in sustainable decision-making by quantitatively assessing the impacts across the entire life cycle.[140] Although LCA is widely employed in biorefineries, research on the LCA of ADESs in biomass pretreatment is still limited. The outcomes of LCA are classified into nine components, encompassing chemical inputs, electricity, transportation, waste incineration, wastewater treatment, ash landfill, direct process discharge, and other inputs.[141]

Currently, DESs are primarily at the laboratory research stage, and for a more efficient and rapid transition to commercialization, it is crucial to conduct comprehensive environmental impact assessments throughout the production, synthesis, and application processes of DESs. A study conducted by Zaib *et al.* employed LCA to evaluate the potential environmental impacts associated with the production of ChCl : U DES.[142] The assessment focused on seven key indicators, including global warming potential (GWP), freshwater eutrophication potential (FEP), terrestrial acidification potential (TAP), metal depletion potential (MDP), water depletion potential (WDP), freshwater ecotoxicity potential (FETP), and human potential toxicity (HTP) (Fig. 17). Comparing ChCl : U DES with four commonly used organic solvents—methanol, ethanol, dichloromethane (DCM), and ethyl acetate, it was found that ChCl : U DES exhibited higher toxicity levels than the other three solvents (methanol, ethanol, and ethyl acetate), except for DCM. Notably, DCM has the highest impact for GWP, TAP, and HTP, with a relative impact value of 100%, and ethyl acetate has the highest relative impact value among FEP, MDP, WDP, and FETP, reaching 100%. The study also examined the influence of four different hydrogen bond donor DESs (ChClEG DES, ChCl : Gly DES, ChCl : CA DES, ChCl : Glu DES) on the life cycle environment of DESs. Among these, ChCl : CA had the most prominent impact across all evaluated metrics, followed by ChCl : Gly and ChCl : Glu. Furthermore, citric acid (CA), which is produced through fermentation, was identified as the most influential component.

However, it is important to note that the production of CA emits 6.02 kg CO_2 per kg, which is 3.7 times higher than the production of urea of equivalent quality. Furthermore, the fermentation process for citric acid production consumes a substantial amount of water (153 L kg^{-1}). Consequently, ChCl : CA DES exhibits the most pronounced environmental impact among the DESs studied. Therefore, adding an appropriate amount of water to DESs to reduce the quantity of DESs used, and consequently lowering the CO_2 emissions, may be a feasible approach. In addition, CO_2 emissions from the operating process are an important indicator of the environmental sustainability of biorefineries, and CO_2 reductions are only meaningful for the conversion of biomass biorefineries into sustainable chemicals. Research in this area is still in its infancy, so it is important to track the carbon footprint and reduce CO_2 emissions while developing low-cost advanced processes (Fig. 18).

The combined use of LCA and TEA allows for a comprehensive assessment of the sustainability of the ADESs production process and helps to formulate rational production strategies and policies for the sustainable development of biomass refining processes.

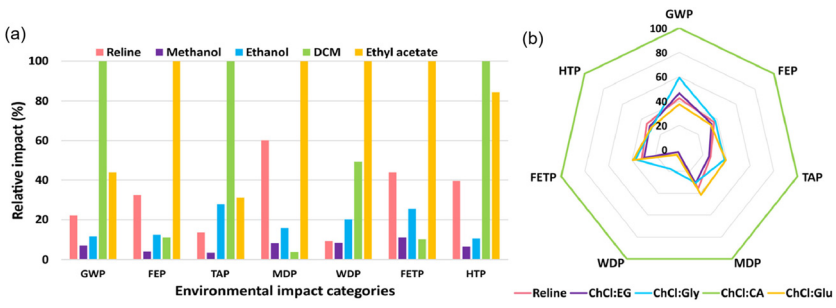


Fig. 17 (a) The life cycle environmental impact of DESs *versus* the conventional organic solvent cradle-to-gate. (b) The life cycle environmental impact of one-kilogram DESs compared to other ChCl-based DESs. Reline: regenerated DES, ChCl : CA DESs only compared to ChCl : EG, ChCl : Gly, and ChCl : Glu DESs (reproduced from ref. 142. Copyright (2022) The Royal Society of Chemistry).

 This journal is © The Royal Society of Chemistry 2023

Published on 16 November 2023. Downloaded by Westlake University on 11/3/2025 9:33:15 AM.

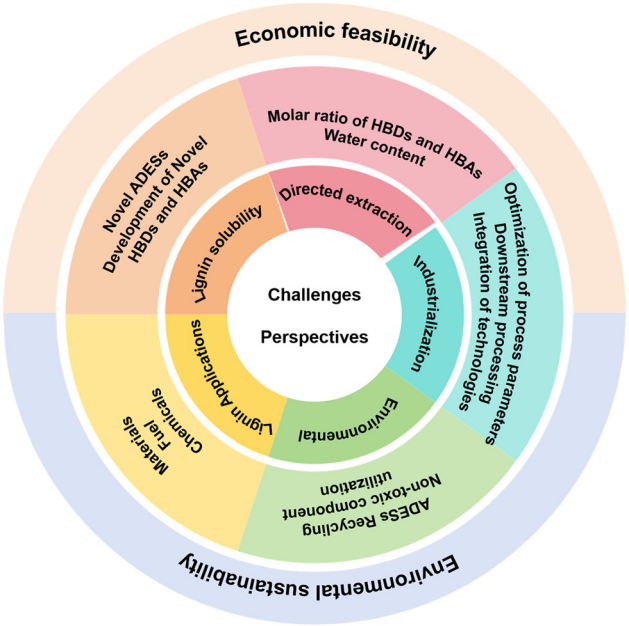


Fig. 18 The challenges and perspectives of ADESs in the field of lignin extraction.

10. Challenges and perspectives

ADESs have gained significant attention as green and sustainable alternatives for lignin extraction. However, some challenges and prospects are still faced in association with the use of ADESs for lignin extraction.

(i) To explore cost-effective and sustainable environment friendly ADESs for enhancing lignin solubility. Lignin is insoluble in many traditional organic solvents, making its extraction difficult. However, certain aqueous DESs, formed by combining HBD and HBA in a specific molar ratio, can solubilize lignin effectively. The tunability of DESs components including water content allows for the optimization of solubility parameters to improve lignin solubility.

(ii) To preserve lignin structure after extraction using ADESs for subsequent valorization. Lignin is a complex highly cross-linked polymer. Traditional lignin extraction methods, such as acidic or alkaline treatments, often lead to structural degradation or modifications. ADESs provide a mild and selective extraction environment, allowing the preservation of lignin's structural integrity. This is particularly important when lignin is intended for value-added applications in industries such as materials, fuels, and chemicals.

(iii) To control the selectivity and preferential extraction of lignin by controlling the HBD and HBA ratio as well as water content. A challenge in lignin extraction is achieving high selectivity toward lignin while minimizing the co-extraction of other components such as hemicellulose and cellulose. ADESs can be tailored to exhibit selectivity toward lignin by adjusting their composition and the extraction conditions.

(iv) To elucidate its environmental sustainability of lignin extraction using ADESs. ADESs offer several environmental advantages compared to traditional organic solvents. They are mostly composed of environmentally benign components, such as sugars, organic acids, or amino acids, making them inherently safe and non-toxic. Additionally, the recyclability of DESs, which allows for their reuse in subsequent extraction cycles, reduces the overall environmental impact of the process.

(v) To develop economically viable and scalable processes. Scalability is a crucial factor for any extraction method aiming for industrial applications. While ADESs for lignin extraction is a relatively young research area, efforts are being made to develop economically viable and scalable processes. The optimization of extraction parameters, downstream processing, and integration with existing biomass conversion technologies are being explored to enable the practical application of ADESs at a larger scale.

11. Conclusions

ADESs has emerged as a promising environmentally friendly solvent for biomass pretreatment, demonstrating significant efficacy in the fractionation and stabilization of lignin. This comprehensive review explores the recent advancements in the application of ADESs for lignin extraction. It covers key aspects such as the impact of water on the performance of DESs, lignin extraction by different ADESs and the mechanistic insights into lignin extraction through ADESs solubilization, the structure and properties of D-lignin, and the promising prospects of high-value applications. Utilizing ADESs represents a significant pathway toward utilizing renewable resources for sustainable energy-driven development. The incorporation of water into DESs not only reduces viscosity and enhances performance but also demonstrates remarkable competitiveness in terms of economic and environmental sustainability. Although notable research progress has been made, further investigations are needed to delve into the molecular-level interactions between ADESs and lignin as well as to assess the economic and environmental aspects of ADESs technology using LCA and TEA. These efforts aim to promote the maturity of ADESs technology, enabling the industrialization of an efficient, cost-effective, and integrated ADESs system for lignocellulose biorefining.

Abbreviations

ADESs	Aqueous deep eutectic solvents
DESs	Deep eutectic solvents
S	Syringyl
H	*p*-Hydroxyphenyl
G	Guaiacyl
HBA	Hydrogen bond acceptor
HBD	Hydrogen bond donor
ChCl	Choline chloride
B	Betaine

Published on 16 November 2023. Downloaded by Westlake University on 11/3/2025 9:33:15 AM.

EG	Ethylene glycol
U	Urea
LA	Lactic acid
EPSR	Empirical potential structure refinement
SDF	Spatial density function
FT-IR	Fourier transform infrared
TGA	Thermogravimetric analysis
PEG	Polyethylene glycol
LCC	Lignin–carbohydrate complex
OA	Oxalic acid
FA	Formic acid
PA	Propionic acid
*p*TSA	*p*-Toluenesulfonic acid monohydrate
MA	Malic acid
P	Proline
Lys	Lysine
Arg	Arginine
His	Histidine
DLS	Dynamic light scattering
p-SA	*p*-Toluene sulfonic acid
L-Arg	L-Arginine
L-Pro	L-Proline
Gly	Glycerol
BDO	1,4-Butanediol
PPD	1,2-Propanediol
OPF	Oil palm fronds
PCA	*p*-Coumaric acid
CA	Choline alanine
CT	Choline threonine
CL	Choline lysine
FF	Furfural
LBHK	Lignin-bound Hibbert ketone
PPE	2-Phenoxy-1-phenylethanol
K–T	Kamlet–Taft
D-lignin	ADES regenerated lignin
GPC	Gel permeation chromatography
XPS	X-ray photoelectron spectroscopy
DSC	Differential scanning calorimetry
PDI	Polydispersity indices
SEM	Scanning electron microscopy
2D-HSQC NMR	Two-dimensional heteronuclear singular quantum correlation nuclear magnetic resonance
LNPs	Lignin nanospheres
MEA	Monoethanolamine
PVA	Polyvinyl alcohol
UV	Ultraviolet
WPU	Waterborne polyurethane
MD	Molecular dynamics
AVM	Avermectin
LXNPs	Lignin–xylan hybrid nanoparticles
NVCL	*N*-Vinyl caprolactam
LCA	Life cycle assessment
TEA	Techno-economic assessment
GWP	Global warming potential
FEP	Freshwater eutrophication potential
TAP	Terrestrial acidification potential
MDP	Metal depletion potential
WDP	Water depletion potential
FETP	Freshwater ecotoxicity potential
HTP	Human potential toxicity
DCM	Dichloromethane
ChOAc	Choline acetate
LevA	Levulinic acid
DFT	Density functional theory
BTEAC	Benzyl triethylammonium chloride
Glu	Glucose
M	Maltose
F	Fructose
X	Xylose
S	Sucrose
Xl	Xylitol
βA	β-Alanine
Sb	Sorbitol

Author contributions

Mingyang Hu: writing – original draft. Yanyan Yu: draw the graphical abstract. Xiaoyan Li and Xinyu Wang: collection of the data. Yun Liu: conceptualization, supervision, writing – review & editing, funding acquisition, project administration.

Conflicts of interest

The authors declare no known competing financial interests.

Acknowledgements

This work was financially funded by the National Key Research & Development Plan Project (grant no. 2022YFB4201801).

References

1 M. D. M. Contreras-Gámez, Á. Galán-Martín, N. Seixas, A. M. C. Lopes, A. Silvestre and E. Castro, *Bioresour. Technol.*, 2023, **369**, 128396.
2 S. Behera, R. Arora, N. Nandhagopal and S. Kumar, *Renewable Sustainable Energy Rev.*, 2014, **36**, 91–106.
3 H. Chen, J. Liu, X. Chang, D. Chen, Y. Xue, P. Liu, H. Lin and S. Han, *Fuel Process. Technol.*, 2017, **160**, 196–206.
4 R. K. Rathour, M. Behl, K. Dhashmana, D. Sakhuja, H. Ghai, N. Sharma, K. R. Meena, A. K. Bhatt and R. K. Bhatia, *Ind. Crops Prod.*, 2023, **204**, 117220.
5 M. Hu, L. Yuan, Z. Cai, J. Zhang, D. Ji and L. Zang, *ACS Omega*, 2021, **6**, 31689–31698.
6 L. A. Z. Torres, A. L. Woiciechowski, V. O. D. A. Tanobe, S. G. Karp and C. R. Soccol, *J. Cleaner Prod.*, 2020, **263**, 121499.

 This journal is © The Royal Society of Chemistry 2023

Published on 16 November 2023. Downloaded by Westlake University on 11/3/2025 9:33:15 AM.

7 C. Li, X. Zhao, A. Wang, G. W. Huber and T. Zhang, *Chem. Rev.*, 2015, **115**, 11559–11624.
8 J. J. Liao, N. H. A. Latif, D. Trache, N. Brosse and M. H. Hussin, *Int. J. Biol. Macromol.*, 2020, **162**, 985–1024.
9 R. Zhao, D. Pei, P. Yu, J. Wei, N. Wang, D. Di and Y. Liu, *J. Sep. Sci.*, 2020, **43**, 348–359.
10 Y. Liu, W. Chen, Q. Xia, B. Guo, Q. Wang, S. Liu, Y. Liu, J. Li and H. Yu, *ChemSusChem*, 2017, **10**, 1857.
11 E. L. Smith, A. P. Abbott and K. S. Ryder, *Chem. Rev.*, 2014, **114**, 11060–11082.
12 Y. Chen and T. Mu, *Green Energy Environ.*, 2019, **4**, 21.
13 A. P. Abbott, G. Capper, D. L. Davies, R. K. Rasheed and V. Tambyrajah, *Chem. Commun.*, 2003, 70–71.
14 Z. Chen, A. Ragauskas and C. Wan, *Ind. Crops Prod.*, 2020, **147**, 112241.
15 S. Hong, X. J. Shen, Z. Xue, Z. Sun and T. Q. Yuan, *Green Chem.*, 2020, **22**, 7219.
16 M. Zhou, O. A. Fakayode, A. E. A. Yagoub, Q. Ji and C. Zhou, *Renewable Sustainable Energy Rev.*, 2022, **156**, 111986.
17 V. Sharma, M. Tsai, C. Chen, P. Sun, A. K. Patel, R. R. Singhania, P. Nargotra and C. Dong, *Bioresour. Technol.*, 2022, **360**, 127631.
18 E. K. New, T. Y. Wu, C. B. T. L. Lee, Z. Y. Poon, Y. L. Loow, L. Y. W. Foo, A. Procentese, L. F. Siow, W. H. Teoh and N. N. N. Daud, *Process Saf. Environ.*, 2019, **123**, 190–198.
19 J. Xie, J. Xu, Z. Zhang, B. Wang, S. Zhu, J. Li and K. Chen, *Chem. Eng. J.*, 2023, **451**, 138591.
20 S. Rozas, C. Benito, R. Alcalde, M. Atilhan and S. Aparicio, *J. Mol. Liq.*, 2021, **344**, 117717.
21 I. M. Marrucho, C. Florindo and L. C. Branco, *ChemSusChem*, 2019, **12**, 1549–1559.
22 F. Gabriele, M. Chiarini, R. Germani, M. Tiecco and N. Spreti, *J. Mol. Liq.*, 2019, **291**, 111301.
23 K. A. Omar and R. Sadeghi, *J. Mol. Liq.*, 2022, **360**, 119524.
24 C. Ma, A. Laaksonen, C. Liu, X. Lu and X. Ji, *Chem. Soc. Rev.*, 2018, **47**, 8685–8720.
25 A. Yadav and S. Pandey, *J. Chem. Eng. Data*, 2014, **59**, 2221–2229.
26 C. Du, B. Zhao, X.-B. Chen, N. Birbilis and H. Yang, *Sci. Rep.*, 2016, **6**, 29225.
27 H. Li, X. Li, D. Li, J. Zhang, H. Nawaz, T. You and F. Xu, *Bioresour. Technol.*, 2022, **351**, 126993.
28 W. Wei and C. Djla, *Bioresour. Technol.*, 2021, **339**, 125587.
29 R. Alcalde, A. Gutiérrez, M. Atilhan and S. Aparicio, *J. Mol. Liq.*, 2019, **290**, 110916.
30 W. Tang, Y. An and K. H. Row, *TrAC, Trends Anal. Chem.*, 2021, **136**, 116187.
31 R. Alcalde, A. Gutiérrez, M. Atilhan and S. Aparicio, *J. Mol. Liq.*, 2019, **290**, 110916.
32 V. Sharma, M.-L. Tsai, C.-W. Chen, P.-P. Sun, A. K. Patel, R. R. Singhania, P. Nargotra and C.-D. Dong, *Bioresour. Technol.*, 2022, **360**, 127631.
33 E. L. Smith, A. P. Abbott and K. S. Ryder, *Chem. Rev.*, 2014, **114**, 11060–11082.
34 H. Qin, X. Hu, J. Wang, H. Cheng, L. Chen and Z. Qi, *Green Energy Environ.*, 2020, **5**, 14.
35 S. Rozas, C. Benito, R. Alcalde, M. Atilhan and S. Aparicio, *J. Mol. Liq.*, 2021, **344**, 117717.
36 O. S. Hammond, D. T. Bowron and K. J. Edler, *Green Chem.*, 2016, **18**, 2736–2744.
37 R. Stefanovic, M. Ludwig, G. B. Webber, R. Atkin and A. J. Page, *Phys. Chem. Chem. Phys.*, 2017, **19**, 3297–3306.
38 O. S. Hammond, D. T. Bowron and K. J. Edler, *Angew. Chem., Int. Ed.*, 2017, **56**, 9782–9785.
39 L. Sapir and D. Harries, *J. Chem. Theory Comput.*, 2020, **16**, 3335–3342.
40 Y. Xu, Y. H. Liu, L. H. Xu, Y. T. He, J. L. Wen and T. Q. Yuan, *Bioresour. Technol.*, 2023, **380**, 129090.
41 V. Sharma, M. L. Tsai, C. W. Chen, P. P. Sun, A. K. Patel, R. R. Singhania, P. Nargotra and C. D. Dong, *Bioresour. Technol.*, 2022, **360**, 127631.
42 V. R. Chourasia, M. Bisht, K. K. Pant and R. J. Henry, *Bioresour. Technol.*, 2022, **351**, 127005.
43 D. Tian, Y. Guo, J. Hu, G. Yang and F. Shen, *Int. J. Biol. Macromol.*, 2019, **142**, 288–297.
44 J. Wang, Y. Wang, Z. Ma and L. Yan, *Green Energy Environ.*, 2020, **5**, 232–239.
45 Q. Zhang, D. O. V. Karine, S. Royer and F. Jérôme, *Chem. Soc. Rev.*, 2012, **41**, 7108–7146.
46 A. K. Kumar, B. S. Parikh and M. Pravakar, *Environ. Sci. Pollut. Res. Int.*, 2016, **23**, 9265–9275.
47 B. Soares, D. J. P. Tavares, J. L. Amaral, A. J. D. Silvestre, C. S. R. Freire and J. O. A. P. Coutinho, *ACS Sustainable Chem. Eng.*, 2017, **5**, 4056–4065.
48 Y. Zhang, H. Ren, H. Maarof, S. M. Udin, Y. Liu, M. Li, H. Alias and E. Duan, *J. Mol. Liq.*, 2023, **374**, 121271.
49 B. Soares, A. M. d. Lopes, P. C. R. P. Armando, J. D. Pinto, C. S. R. Freire and J. A. P. Coutinho, *Ind. Crops Prod.*, 2021, **160**, 113128.
50 Y. Liang, W. Duan, X. An, Y. Qiao, Y. Tian and H. Zhou, *Bioresour. Technol.*, 2020, **310**, 123389.
51 M. P. Gundupalli, K. Cheenkachorn, S. Chuetor, S. Kirdponpattara, S. P. Gundupalli, P.-L. Show and M. Sriariyanun, *Carbohydr. Polym.*, 2023, **306**, 120599.
52 S. Balasubramanian and P. Venkatachalam, *Biomass Bioenergy*, 2023, **173**, 106776.
53 J. Cui, R. Chen, L. Lei and Y. Hou, *Biomass Convers. Biorefin.*, 2022, DOI: **10.1007/s13399-022-03498-7**.
54 X. D. Hou, G. J. Feng, M. Ye, C. M. Huang and Y. Zhang, *Bioresour. Technol.*, 2017, **238**, 139–146.
55 Z. Zhao, X. Chen, M. F. Ali, A. A. Abdeltawab, S. M. Yakout and G. Yu, *Bioresour. Technol.*, 2018, **263**, 325–333.
56 J. Cheng, C. Huang, Y. Zhan, X. Liu, J. Wang, X. Meng, C. G. Yoo, G. Fang and A. J. Ragauskas, *Green Chem.*, 2023, **25**, 1571–1581.
57 C. Zhu, B. Xianglan, A. Lusi and W. Caixia, *ACS Sustainable Chem. Eng.*, 2018, **6**, 12205–12216.
58 M. Shaibuna, L. V. Theresa and K. Sreekumar, *Soft Matter*, 2022, **18**, 2695–2721.
59 Y. Zhu, Z. Li and J. Chen, *Green Energy Environ.*, 2019, **4**, 210–244.

Published on 16 November 2023. Downloaded by Westlake University on 11/3/2025 9:33:15 AM.

60 Z. Zhou, D. Liu and X. Zhao, *Renewable Sustainable Energy Rev.*, 2021, **146**, 111169.
61 A. R. Mankar, A. Pandey, A. Modak and K. K. Pant, *Bioresour. Technol.*, 2021, **334**, 125235.
62 A. M. d. C. Lopes, J. R. B. Gomes, J. A. P. Coutinho and A. J. D. Silvestre, *Green Chem.*, 2020, **22**, 2474.
63 D. M. Miles-Barrett, A. R. Neal, C. Hand, J. R. D. Montgomery, I. Panovic, O. S. Ojo, C. S. Lancefield, D. B. Cordes, A. M. Z. Slawin, T. Lebl and N. J. Westwood, *Org. Biomol. Chem.*, 2016, **14**, 10023–10030.
64 S. S. Stahl, J. J. Coon, A. Ulbrich and A. Rahimi, *Nature*, 2014, **515**, 249.
65 V. Z. Ong, T. Wu, K. Chu, W. J. Sun and K. Shak, *Ind. Crops Prod.*, 2021, **160**, 112974.
66 T. Li, G. Lyu, Y. Liu, R. Lou, L. A. Lucia, G. Yang, J. Chen and S. Ham, *Int. J. Mol. Sci.*, 2017, **188**, 2266.
67 A. M. C. Lopes, J. R. B. Gomes, J. A. P. Coutinho and A. J. D. Silvestre, *Green Chem.*, 2020, **22**, 2474–2487.
68 M. Hu, L. Yuan, Z. Cai, W. Zhang, Q. Fu and D. Ji, *Bioresour. Technol.*, 2023, **367**, 128242.
69 Z. Chen and C. Wan, *Bioresour. Technol.*, 2018, **250**, 532–537.
70 W. Liu, C. Ning, Z. Li, X. Li, H. Wang and Q. Hou, *Ind. Crops Prod.*, 2023, **194**, 116342.
71 Y. Liu, C. Zhang, Z. Liu, D. G. Truhlar, Y. Wang and X. He, *Nat. Comput. Sci.*, 2023, **3**, 48–58.
72 A. M. d. C. Lopes, J. R. B. Gomes, J. A. P. Coutinho and A. J. D. Silvestre, *Green Chem.*, 2020, **22**, 2427.
73 Q. Xia, Y. Liu, J. Meng, W. Cheng, W. Chen, S. Liu, Y. Liu, J. Li and H. Yu, *Green Chem.*, 2018, **20**, 2711.
74 R. Radhakrishnan, B. Manna and A. Ghosh, *Renewable Energy*, 2023, **206**, 47–59.
75 X. Wu, H. Lian, X. Li and J. Xiao, *Int. J. Biol. Macromol.*, 2023, **253**, 126664.
76 H. Xu, J. Peng, Y. Kong, Y. Liu, Z. Su, B. Li, X. Song, S. Liu and W. Tian, *Bioresour. Technol.*, 2020, **310**, 123416.
77 D. Tian, Y. Guo, J. Hu, G. Yang and F. Shen, *Int. J. Biol. Macromol.*, 2019, **142**, 288–297.
78 J. Yang, W. Zhang, Y. Tang, M. Li, F. Peng and J. Bian, *Bioresour. Technol.*, 2023, **368**, 128258.
79 Z. Chen, W. D. Reznicek and C. Wan, *Bioresour. Technol.*, 2018, **263**, 40–48.
80 H. Yang, C. G. Yoo, X. Meng, Y. Pu, W. Muchero, G. A. Tuskan, T. J. Tschaplinski, A. J. Ragauskas and L. Yao, *Bioresour. Technol.*, 2020, **295**, 122240.
81 A. Brandt, L. Chen, B. E. van Dongen, T. Welton and J. P. Hallett, *Green Chem.*, 2015, **17**, 5019–5034.
82 B. Soares, A. M. C. Lopes, A. J. D. Silvestre, P. C. R. Pinto, C. S. R. Freire and J. A. P. Coutinho, *Ind. Crops Prod.*, 2021, **160**, 113128.
83 Z. Chen, X. Bai, A. Lusi, H. Zhang and C. Wan, *ACS Sustainable Chem. Eng.*, 2020, **8**, 9783–9793.
84 J. Cheng, C. Huang and Y. Zhan, *Chem. Eng. J.*, 2022, **443**, 136395.
85 H. Zhang, H. Zhang, S. Tian and S. Fu, *Bioresour. Technol.*, 2021, **341**, 125848.
86 Y. Chen, L. Zhang, J. Yu, Y. Lu, B. Jiang, Y. Fan and Z. Wang, *R. Soc. Open Sci.*, 2019, **6**, 181757.
87 L. Yang, T. Zheng, C. Huang and J. Yao, *Renewable Energy*, 2022, **195**, 681–687.
88 Q. Ji, X. Yu, P. Wu, A. E.-G. A. Yagoub, L. Chen, A. T. Mustapha and C. Zhou, *Ind. Crops Prod.*, 2021, **173**, 114108.
89 L. Y. Ee, Y. K. Tan, J. Miao, H. T. Chu and S. F. Y. Li, *Green Chem.*, 2023, **25**, 3137–3151.
90 H. Li, X. Li, T. You, D. Li, H. Nawaz, X. Zhang and F. Xu, *Int. J. Biol. Macromol.*, 2021, **193**, 319–327.
91 S. Han, R. Wang, K. Wang, J. Jiang and J. Xu, *Bioresour. Technol.*, 2022, **363**, 127905.
92 Z. Guo, J. Mao, Q. Zhang and F. Xu, *Ind. Crops Prod.*, 2022, **188**, 115453.
93 C. Alvarez-Vasco, R. Ma, M. Quintero, M. Guo, S. Geleynse, K. K. Ramasamy, M. Wolcott and X. Zhang, *Green Chem.*, 2016, **18**, 5133.
94 Y. T. Tan, A. S. M. Chua and G. C. Ngoh, *Bioresour. Technol.*, 2020, **297**, 122522.
95 Y. Song, X. Shi, S. Ma, X. Yang and X. Zhang, *Cellulose*, 2020, **27**, 8301–8315.
96 Z. Wang, S. Hong, J. Wen, C. Ma, L. Tang, H. Jiang, J. Chen, S. Li, X. Shen and T. Yuan, *ACS Sustainable Chem. Eng.*, 2019, **8**, 1050–1057.
97 Y. Ci, F. Yu, C. Zhou, H. Mo, Z. Li, Y. Ma and L. Zang, *Green Chem.*, 2020, **22**, 8713–8720.
98 K. T. Lin, C. Wang, M. F. Guo, E. Aprà, R. Ma, A. J. Ragauskas and X. Zhang, *Proc. Natl. Acad. Sci. U. S. A.*, 2023, **120**, e2307323120.
99 Z. Chen, X. Bai, L. A. A and C. Wan, *ACS Sustainable Chem. Eng.*, 2018, **6**, 12205–12216.
100 E. Melro, A. Riddell, D. Bernin, A. M. R. da Costa, A. J. M. Valente, F. E. Antunes, A. Romano, M. Norgren and B. Medronho, *Biomacromolecules*, 2023, **24**, 3094–3104.
101 Y. Liu, N. Deak, Z. Wang, H. Yu, L. Hameleers, E. Jurak, P. J. Deuss and K. Barta, *Nat. Commun.*, 2021, **12**, 5424.
102 M. Mennani, M. Kasbaji, A. A. Benhamou, A. Boussetta, A. A. Mekkaoui, N. Grimi and A. Moubarik, *Green Chem.*, 2023, **25**, 2896–2929.
103 Q. Ji, X. Yu, P. Wu, E. G. A. Yagoub, L. Chen, A. T. Mustapha and C. Zhou, *Ind. Crops Prod.*, 2021, **173**, 114108.
104 D. A. Ali and M. M. Mehanna, *Int. J. Biol. Macromol.*, 2022, **221**, 934–953.
105 B. Abraham, V. L. Syamnath, K. B. Arun, P. M. F. Zahra, P. Anjusha, A. Kothakotta, Y.-H. Chen, V. K. Ponnusamy and P. Nisha, *Sci Total Environ.*, 2023, **881**, 163316.
106 A. d. E. S. Pereira, J. L. Oliveira, S. M. Savassa, C. B. Rogério, G. A. Medeiros and L. F. Fraceto, *J. Cleaner Prod.*, 2022, **345**, 131145.
107 B. Du, W. Li, Y. Bai, Z. Pan, Q. Wang, X. Wang, H. Ding, G. Lv and J. Zhou, *Int. J. Biol. Macromol.*, 2022, **214**, 170–180.

 This journal is © The Royal Society of Chemistry 2023

Published on 16 November 2023. Downloaded by Westlake University on 11/3/2025 9:33:15 AM.

108 Y. Guo, L. Xu, F. Shen, J. Hu, M. Huang, J. He, Y. Zhang, S. Deng, Q. Li and D. Tian, *Chemosphere*, 2022, **286**, 131798.
109 T. Luo, C. Wang, X. Ji, G. Yang, J. Chen, C. G. Yoo, S. Janaswamy and G. Lyu, *Int. J. Biol. Macromol.*, 2021, **183**, 781–789.
110 H. Zhang, Y. Shi, M. Li, J. Chen, Y. Xin, L. Zhang, Z. Gu, J. Liu and R. Liu, *Chem. Eng. Sci.*, 2022, **256**, 117694.
111 L. H. Xu, C. Y. Ma, C. Zhang, J. Liu, X. P. Peng, S. Q. Yao, D. Y. Min, T. Q. Yuan and J. L. Wen, *Ind. Crops Prod.*, 2022, **176**, 114275.
112 Y. Park, Y. S. Lee, S. Sarker, E. H. Ham and U. T. Lim, *Biol. Control*, 2023, **179**, 105166.
113 L. Wang, X. Li, C. Wan, K. Zhang, Z. Wu, F. Hu, R. Zhang, X. Fu and H. Yu, *Sci. Total Environ.*, 2023, **861**, 161090.
114 M. Yu, D. He, Y. Zhang, D. He, X. Wang and J. Zhou, *Int. J. Biol. Macromol.*, 2021, **192**, 498–505.
115 Y. Li, S. Zhao, X. Song, D. Y. Cao, K. Li, M. Hassanpour and Z. Zhang, *Macromol. Mater. Eng.*, 2021, **307**, 2100628.
116 X. Li, J. Xiao, W. Wei, M. Yan, D. Guo and H. Lian, *Int. J. Biol. Macromol.*, 2023, **225**, 1405–1414.
117 Y. Mukheja, J. Kaur, K. Pathania, S. P. Sah, D. B. Salunke, A. T. Sangamwar and S. V. Pawar, *Int. J. Biol. Macromol.*, 2023, **241**, 124601.
118 M. Stanisz, Ł. Klapiszewski, M. N. Collins and T. Jesionowski, *Mater. Today Chem.*, 2022, **26**, 101198.
119 F. Shen, S. Wu, M. Huang, L. Zhao, J. He, Y. Zhang, S. Deng, J. Hu, D. Tian and F. Shen, *Green Chem.*, 2022, **24**, 5242.
120 Y. Jiang, Y. Chen, D. Tian, F. Shen and F. Shen, *Soft Matter*, 2020, **16**, 9083.
121 R. Liu, T. Ding, P. Deng, X. Yan, F. Xiong, J. Chen and Z. Wu, *Biomater. Sci.*, 2022, **10**, 4284–4292.
122 H. Poy, E. Lladosa, A. Arcís, C. Gabaldón and S. Loras, *Ind. Crops Prod.*, 2023, **206**, 117639.
123 G. Wang, M. Huang, F. Li, Q. Li, F. Chen, S. Wang, Z. Ling and Z. Ji, *Int. J. Biol. Macromol.*, 2023, **254**, 127673.
124 Y. Q. Sun, Y. Yuan, K. X. Dai and Z. L. Xiu, *Bioresour. Technol.*, 2023, **384**, 129289.
125 X.-J. Shen, J.-L. Wen, Q.-Q. Mei, X. Chen, D. Sun, T.-Q. Yuan and R.-C. Sun, *Green Chem.*, 2019, **21**, 275–283.
126 Y. Huang, Y. Xu, Y. Zhu, R. Huang, Y. Kuang, J. Wang, W. Xiao, J. Lin and Z. Liu, *Bioresour. Technol.*, 2022, **366**, 128186.
127 Y. Bai, X.-F. Zhang, Z. Wang, T. Zheng and J. Yao, *Bioresour. Technol.*, 2022, **347**, 126723.
128 X. Diao, N. Ji, X. Li, Y. Rong, Y. Zhao, X. Lu, C. Song, C. Liu, G. Chen and L. Ma, *Appl. Catal., B*, 2022, **305**, 121067.
129 J. Sunthornvarabhas, S. Liengprayoon and T. Suwonsichon, *Ind. Crops Prod.*, 2017, **109**, 857–861.
130 Y. Xi, S. Huang, D. Yang, X. Qiu, H. Su, C. Yi and Q. Li, *Green Chem.*, 2020, **22**, 4321–4330.
131 A. Y. Gebreyohannes, S. L. Aristizábal, L. Silva, E. A. Qasem, S. Chisca, L. Upadhyaya, D. Althobaiti, J. A. P. Coutinho and S. P. Nunes, *Green Chem.*, 2023, **25**, 4769–4780.
132 A. U. R. Akond and J. G. Lynam, *Case Stud. Constr. Mater.*, 2020, **13**, e00460.
133 B. Xue, Y. Yang, R. Tang, D. Xue and X. Li, *Int. J. Biol. Macromol.*, 2020, **164**, 480–488.
134 M. K. Awasthi, S. Sarsaiya, A. Patel, A. Juneja, R. P. Singh, B. Yan, S. K. Awasthi, A. Jain, T. Liu, Y. Duan, A. Pandey, Z. Zhang and M. J. Taherzadeh, *Renewable Sustainable Energy Rev.*, 2020, **127**, 109876.
135 M. Del Mar Contreras-Gámez, Á. Galán-Martín, N. Seixas, A. M. da Costa Lopes, A. Silvestre and E. Castro, *Bioresour. Technol.*, 2023, **369**, 128396.
136 W. Li, X. Tan, C. Miao, Z. Zhang, A. Ragauskas, Y. Wang and X. Zhuang, *Green Chem.*, 2023, **25**, 1169–1178.
137 R. T. Elander, B. E. Dale, M. Holtzapple, M. R. Ladisch, Y. Y. Lee, C. Mitchinson, J. N. Saddler and C. E. Wyman, *Cellulose*, 2009, **16**, 649–659.
138 G. C. Dugoni, A. Mezzetta, L. Guazzelli, C. Chiappe, M. Ferro and A. Mele, *Green Chem.*, 2020, **22**, 8680–8691.
139 L. Chen, Q. Yu, Q. Wang, W. Wang, W. Qi, X. Zhuang, Z. Wang and Z. Yuan, *Cellulose*, 2019, **26**, 1947–1959.
140 M. del Mar Contreras-Gámez, Á. Galán-Martín, N. Seixas, A. M. da Costa Lopes, A. Silvestre and E. Castro, *Bioresour. Technol.*, 2023, **369**, 128396.
141 F. Liu, X. Dong, X. Zhao and L. Wang, *Energy Convers. Manage.*, 2021, **246**, 114653.
142 Q. Zaib, M. J. Eckelman, Y. Yang and D. Kyung, *Green Chem.*, 2022, **24**, 7924–7930.
143 M. S. C. Zain, J. X. Yeoh, S. Y. Lee and K. Shaari, *Sustainability*, 2021, **13**, 12981.
144 Y. Dai, J. Van Spronsen, G. J. Witkamp, R. Verpoorte and Y. H. Choi, *Anal. Chim. Acta*, 2013, **766**, 61–68.
145 A. L. Li, X. Hou, K. Lin, X. Zhang and M. Fu, *J. Biosci. Bioeng.*, 2018, **126**, 346–354.
146 C. B. T. L. Lee, T. Y. Wu, K. J. Yong, C. K. Cheng, L. F. Siow and J. M. Jahim, *Sci. Total Environ.*, 2022, **827**, 154049.
147 Y. Wang, W. J. Zhang, J. Y. Yang, M. F. Li, F. Peng and J. Bian, *Bioresour. Technol.*, 2022, **354**, 127139.
148 Y. Zhu, B. Qi, X. Liang, J. Luo and Y. Wan, *Polymers*, 2021, **13**, 1170.
149 X. Chen, J. Jiang, J. Zhu, W. Song, C. Liu and L.-P. Xiao, *Bioresour. Technol.*, 2022, **362**, 127788.
150 L. Chen, Q. Yu, Q. Wang, W. Wang, W. Qi, X. Zhuang, Z. Wang and Z. Yuan, *Cellulose*, 2019, **26**, 1947–1959.
151 R. Lou, R. Ma, K. T. Lin, A. Ahamed and X. Zhang, *ACS Sustainable Chem. Eng.*, 2019, **7**, 10248–10256.
152 K. Nie, S. Liu, T. Zhao, Z. Tan, Y. Zhang, Y. Song, B. Li, L. Li, W. Lv and G. Han, *Biomass Convers. Biorefin*, 2022, DOI: **10.1007/s13399-022-03496-9**.
153 H. Zhang, Y. Shi, M. Li, J. Chen, Y. Xin, L. Zhang, Z. Gu, J. Liu and R. Liu, *Chem. Eng. Sci.*, 2022, **256**, 117649.
154 Z. Yan, Z. Wang, Y. Chen, C. Liu, Y. Liu, R. Li, M. Si and Y. Shi, *Biotechnol. Bioeng.*, 2023, **120**, 1557–1568.

155 C. Zhang, K. N. Guo, C.-Y. Ma, J. Bian, J. L. Wen and T. Q. Yuan, *Ind. Crops Prod.*, 2023, **193**, 116204.
156 W. Zhang, J. Shen, P. Gao, Q. Jiang and W. Xia, *Ind. Crops Prod.*, 2022, **188**, 115651.
157 W. H. Gong, C. Zhang, J. W. He, Y. Y. Gao, Y. J. Li, M. Q. Zhu and J. L. Wen, *Int. J. Biol. Macromol.*, 2022, **209**, 188–197.
158 J. Xie, J. Chen, Z. Cheng, S. Zhu and J. Xu, *Carbohydr. Polym.*, 2021, **269**, 118321.
159 H. Li, J. Liang, L. Chen, M. Ren and C. Zhou, *Ind. Crops Prod.*, 2023, **192**, 116034.
160 C. Ma, X. Peng, S. Sun, J. Wen and T. Yuan, *Int. J. Biol. Macromol.*, 2021, **192**, 417–425.

Published on 16 November 2023. Downloaded by Westlake University on 11/3/2025 9:33:15 AM.

This journal is © The Royal Society of Chemistry 2023